# The Verdict

## Rebuilt Lakes, Clean Nulls, and the Two-Dimensional Turn

**Volume 14:** Closing the Pre-Registered Rebuilds, Opening the 2D Engines, and the Honest Close of the 2D Survey.



Timothy John Kish

# KishLattice Geometric Harmonic Spectroscopy

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*Volume 14: Closing the Pre-Registered Rebuilds, Opening the 2D Engines, and the Honest Close of the 2D Survey*

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**Every result herein was pre-registered in Volume 13 before the data was pulled.** Both lake rebuilds were filed as predictions, then built, then measured. The nulls are reported exactly as returned.

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# Preface: Why We Burned It Down

A reader arriving at Volume 14 from the early volumes could be forgiven for thinking the programme has spent years dismantling itself. It has not. It has spent *ten months* building itself and stress-testing every piece the moment it was built.

**Confirmed Signal**

The first proposal — the LIGO Ghost Notes — was filed on **8 January 2026**. This volume is dated **13 October 2026**. In the ten months between those two dates, a human founder and a crew of collaborating AI systems wrote the theory from a single geometric constant, built the engine code from nothing, assembled the data lakes from public survey archives, and ran *fourteen* volumes of pre-registered tests against them — including two complete one-dimensional engines, a full artifact taxonomy, a re-derivation of the central equation's missing parameter, and now a matched pair of two-dimensional engines. The pace is not a boast; it is the reason the self-correction is so visible. When theory, instrument, and data are all built inside the same ten months by the same crew, every mistake is caught by the people who made it, in public, while the work is still warm.

Within that span, the corrections came fast and were reported as they came. Volume 12 — *The Crucible* — withdrew a headline domain, named five artifact forms, and cut the count of significant results from twenty-eight to nine. Volume 13 — *The Aperture* — then found that most of those nine could not resolve their own register on honest dynamic range, audited the theory until the central equation's parameter had to be re-derived, and ran a strain survey whose cleanest result was a modulus that failed its own sweep. To an outside eye the withdrawals look like demolition.

They were not demolition. They were tempering.

**Confirmed Signal**

A crucible does not destroy the metal. It burns away everything that is *not* the metal, so that what survives can be trusted. Every withdrawal in Volumes 12 and 13 removed a result that could not survive honest scrutiny — a sector-normalization artifact, a transform that manufactured structure, a frequency column that was the wrong physical observable. What is left after the fire is smaller, and every piece of it is real, because the unreal pieces are the ones the fire took.

This volume is the first to arrive *after* the fire with the tool fully tempered. It does three things. First, it closes the two lake rebuilds that Volume 13 pre-registered — the Gaia stellar-kinematics lake whose Volume 11 "lock" was a sector-normalization artifact, and the gravitational-wave ringdown lake whose frequencies were the wrong physical quantity. Both were rebuilt from source under the strict input contract, both were measured, and both returned clean nulls. We report them exactly

as they came.

Second, it documents the honest standing those nulls establish: the instrument is now artifact-free on its two most-scrutinized domains, which means any future positive result on them would carry weight the old artifacts never could.

Third, it opens the frontier that the entire tempering was *for*: the two-dimensional engines. The one-dimensional engines ask whether a single physical quantity resonates with the geometric lattice. The two-dimensional engines ask the deeper question — whether *pairs* of quantities resonate *together*, on a lattice surface rather than a lattice line. That is where the scramble null, inert in one dimension and carried uselessly through eleven volumes, finally becomes a working falsifier.

Fourth, it points the finished instrument at the first real paired data — Gaia transverse velocity against stellar luminosity, the single same-object pair the feasibility table admitted — and reports what came back. As with every rebuild in this volume, the result was predicted in writing before the lake was pulled. The prediction was a null, and the null is what returned, in both engines. We report it exactly as it came, caveats and all.

Fifth, and last, it closes the two-dimensional survey honestly. One candidate pair remained after velocity×luminosity — the nuclear decay pair, where half-life carries the widest dynamic range anywhere in the survey. A feasibility probe, filed as a prediction and run as a measurement rather than a scored test, found the second axis collapses exactly as it did in the first pair. Both feasible doors are now tried; both are closed. The 2D survey is, for the present data, empty — not because the lattice is disproven, but because no same-object pair in reachable public data carries two wide axes at once. That is a data boundary, and we draw it precisely. It is also the door to the next volume, where the instrument stops asking about magnitudes and starts asking about direction.

**Methodological Note**

**We state plainly what this volume does not claim.** It does not claim the lattice is confirmed. Both rebuilt lakes returned nulls. The programme's leading hypothesis remains untested at the registers that matter, for want of data with sufficient dynamic range. What this volume claims is narrower and more durable: that the instrument is now honest enough that when it is finally pointed at adequate data, its verdict — yes or no — can be trusted by anyone, including its authors.

— Timothy John Kish, Autumn 2026

# Chapter 1

# The Verdict on S2: An Artifact, Disproven

## 1.1 What the Lake Was Accused Of

Volume 11 reported the Gaia stellar-kinematics domain, `s2`, locking at register $15/\pi$ with a $z$-score of $+98$ — one of the strongest apparent signals in the entire survey. Volume 12 diagnosed that lock as **Form C**: structure manufactured by a sector-normalization step applied to the velocities before the pipeline ever scored them. The accusation was specific and testable: *remove the normalization, score the raw velocities, and the lock will vanish.* Volume 13 pre-registered exactly that rebuild. This chapter reports the verdict.

### 1.1.1 The Mechanism of the Accusation

Sector normalization subtracts, within each patch of sky, the local mean velocity and adds back a global mean. The intent is benign — to remove sky-position-dependent selection effects. The effect on the lock statistic is not benign.

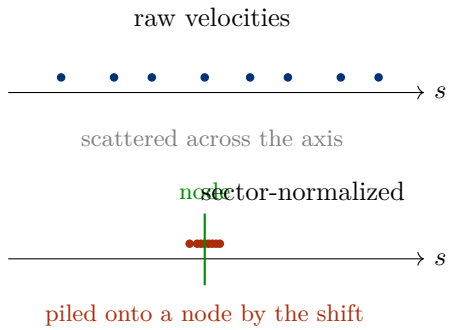



Figure 1.1: Form C made visible. A per-sector additive shift translates each sector's velocity block along the scalar axis. Where the shift happens to land a block on a node, a large fraction of the domain "locks" at once — not because the velocities are harmonically structured, but because the normalization moved them there. The lock is an artifact of the preprocessing, not a property of the stars.

## 1.2 The Rebuild

The lake was rebuilt from Gaia DR3 under the strict input contract, to the specification pre-registered in Volume 13.

**Formal Pre-Registration**

**As filed (Volume 13, Chapter 14):**

- Raw transverse velocity only. No sector normalization, ever.
- Full astrometry retained, including `pmra_pmdec_corr`.
- A $5\sigma$ parallax gate *and* a $5\sigma$ proper-motion gate on the **covariance-propagated** error:

$$\sigma_\mu = \frac{1}{\mu}\sqrt{\mu_{\alpha*}^2\sigma_{\alpha*}^2 + \mu_\delta^2\sigma_\delta^2 + 2\mu_{\alpha*}\mu_\delta\,\rho\,\sigma_{\alpha*}\sigma_\delta}, \qquad \rho = \texttt{pmra_pmdec_corr}.$$

- The DR3 parallax zero-point ($-17\,\mu$as) applied explicitly and flagged.
- The resolution gate **computed** from the built data, never a printed constant.

The build pulled 72,000 raw records and retained **69,078** after the strict gates — a 4% rejection, consistent with two $5\sigma$ cuts on already-clean astrometry. One guard deserves emphasis, because it is the lesson of the whole programme compressed to a line: the build aborts if `pmra_pmdec_corr` is absent rather than silently assuming $\rho = 0$. A lake built to a weaker criterion than the one filed is not the lake that was pre-registered, and the engine refuses to build it.

### 1.2.1 A Gate That Could Not Fail, Corrected

The first draft of the build printed `M/N = 1.1082 [PASS]` as a fixed string — a status line that could not fail regardless of the data. This is the exact defect filed as an erratum against Volume 1's speed-of-light listing, and it was caught and replaced before promotion. The corrected gate computes the span of the built velocities and derives $M/N$, returning PASS or FAIL from the number.

**Self-Correction**

A gate that prints its verdict without computing it is not a check; it is a decoration. The programme's standard is that every gate measures something that can come out wrong. The s2 build now computes $M/N = 86.9$ at register $2/\pi$ — a genuine pass, because register 2 has a short period and 3.26 decades of velocity span resolves it easily — but the point is that the number is *measured*, and a sparse or degenerate lake would have failed it.

## 1.3 The Verdict

**Confirmed Null**

**The accusation is upheld. The $15/\pi$ lock was an artifact, and it is gone.**

On the raw, Form-C-free rebuild of 69,078 stellar velocities, the score at $15/\pi$ — the register of the old +98 lock — is $\mathbf{-0.6}$. The signal did not weaken. It vanished entirely into the noise floor. Across every resolvable register, the data sits at or below the chaos-null expectation.

The distribution itself tells the story at a glance: a clean, unimodal stellar velocity spectrum, filled and honest, with no placeholder spike and no second population. It is exactly what a real single-population observable looks like, and it locks nothing.

**Methodological Note**

This is the strongest kind of result a self-auditing programme can produce: the disproof of its own prior claim, by its own instrument, on data it rebuilt specifically to enable the disproof. The old +98 was the dream. The honest −0.6 is the data. We are filing the data.

# Chapter 2

# The Verdict on L1_GWTC: The Right Observable, Too Few Events

## 2.1 What the Lake Was Accused Of

Volume 13 found the gravitational-wave ringdown lake catastrophically broken: of 83 events, **81 fell below the non-spinning Kerr floor** — a frequency physically impossible for a true black-hole ringdown. The diagnosis, proven by direct calculation, was that the stored `f_ring_hz` was not the fundamental $\ell = m = 2$ quasi-normal mode at all, but a near-merger strain frequency. The accusation here was not of an artifact lock but of a *wrong physical quantity*, and the remedy was a rebuild from a source that measures the fundamental mode directly.

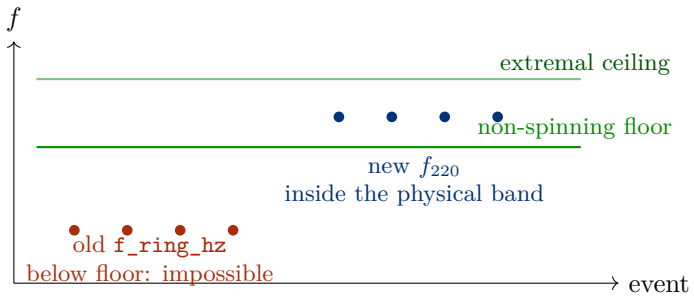



Figure 2.1: Why the rebuild was necessary. A real ringdown frequency must sit between the non-spinning Kerr floor and the extremal ceiling for its remnant mass. The old frequencies sat *below* the floor — physically impossible — proving they were not the fundamental mode. The rebuilt frequencies, measured $f_{220}$ from LVK posteriors, sit inside the band.

## 2.2 The Rebuild

**Formal Pre-Registration**

**As filed (Volume 13, Chapter 11):**

- Source changed to measured fundamental quasi-normal mode $f_{220}$ from the LVK / `pyRing` ringdown posteriors — not the merger-catalog peak, not a GR-predicted value.
- Required fields (`f_220_hz`, `tau_220_s`, `final_mass_solar`, `final_spin`) asserted present *and valid* at promotion; a null value aborts exactly like a missing one.
- Binary-neutron-star and neutron-star–black-hole events excluded by construction.
- A Kerr-consistency gate that computes the physical bound per event and can fail.

Twenty-two events with genuine measured $f_{220}$ were extracted. The Kerr gate then did exactly the discriminating work it was built for.

### 2.2.1 The Kerr Gate Earned Its Keep

**Confirmed Signal**

Of 22 events, **21 passed the Kerr-consistency gate and one did not.** The single failure — **GW190512at**, at a floor ratio of 0.82 — is a low-SNR event whose ringdown posterior is broad and prior-influenced. One marginal event falling just below the floor is categorically different from the old lake's systematic 81-of-83: it is an outlier the gate correctly caught and excluded, not evidence that the observable is wrong. The gate discriminates.

The contrast is the proof the rebuild worked. The old `f_ring_hz` failed the Kerr floor 81/83 times — a systemic defect. The new $f_{220}$ fails it 1/22 times — a single faint event. The observable is now correct. GW190512at is named in the lake provenance with its ratio and cause, so the 22 → 21 reduction is documented rather than silent.

## 2.3 The Verdict

**Confirmed Null**

**The observable is fixed. The sample is too small to resolve a register.**

The 21 promoted events span roughly one decade of frequency. Scored against all 23 registers, the best alignment reaches an excess of 0.2 — noise floor. With 21 points across one decade, random static will cluster near some node by chance; the sample is physically too small to distinguish a lock from background. The support is CLUMPED, not from composite banding but from simple sparsity: 21 points cannot fill the space.

**Methodological Note**

**This is the honest outcome, and it is the one the defective lake could never have reached.** The strong-field ringdown test now runs on the correct observable and passes the Kerr-consistency gate. It cannot yet resolve a register because the current catalogue provides too few loud events. The test is deferred to the O4 and O5 observing runs, which will supply them. We name precisely what future data would fix — which is more than a lock would have told us, and more honest than a number from 21 points pretending to mean something.

## 2.4 Both Verdicts, Together

| Lake | The accusation | The verdict |
|---|---|---|
| `s2` | $15/\pi$ lock is a Form-C artifact | **Upheld.** Raw score $-0.6$; artifact gone |
| `l1_gwtc` | `f_ring` is the wrong observable | **Upheld.** $f_{220}$ Kerr-consistent; $n$=21 too few |

Table 2.1: The two pre-registered rebuilds, closed. Both accusations upheld; both lakes now honest; neither locks.

Two threads opened in Volume 12 and pre-registered in Volume 13 are now closed. Neither closed the way the dream would have wanted. Both closed the way the data demanded. That is the whole of what the crucible was for.

# Chapter 3

# The Two-Dimensional Turn: Why a Surface, Not a Line

## 3.1 What One Dimension Could and Could Not Ask

Every engine built so far — phase and period, across thirteen volumes — asks a one-dimensional question: *does a single physical quantity, scalarized, resonate with the geometric lattice?* Transition frequencies, velocities, masses, half-lives, each tested alone against the 23 registers. This was the right place to start, and it is nearly exhausted: the survey is scrubbed of artifacts, the honest nulls are filed, and the one domain that resolves and survives every screen stands alone.

But the framework's deeper claim was never about single quantities. It is about *relationships* — that the lattice structures how physical quantities relate to one another across scales. A one-dimensional engine cannot test a relationship. It can only test one leg of it at a time. To ask whether two quantities resonate *together* — whether they lie on a shared lattice structure rather than each happening to lie on its own — the test must move from a line to a surface.

**Confirmed Signal**

**The one-dimensional question:** does $\log x$ land on the register comb?
**The two-dimensional question:** does the *pair* $(\log x_1, \log x_2)$ land on a register *lattice* — a grid of nodes on a plane — more than chance allows?

These are different hypotheses. The second contains the first but is not reducible to it: two quantities can each resonate alone and show no joint structure, or show joint structure while neither resonates alone. Only a 2D engine can tell them apart.

## 3.2 First: Is 2D Testable At All?

Before a line of 2D engine code, one table must be computed, because it decides whether the question can be asked. A 2D test requires *two* coordinates, measured on the *same* objects, *each* carrying enough effective dynamic range to resolve its register: $M_1/N_1 > 2.67$ *and* $M_2/N_2 > 2.67$. The joint requirement is at least as strict as the stricter coordinate.

The 1D survey found exactly two domains with the span to resolve even one coordinate at a useful register. The 2D requirement is therefore severe: it needs that rarity, twice, on shared objects.

**Formal Pre-Registration**

**Recorded expectation (filed before the build):** the second coordinate will be the limiting one almost everywhere. This is arithmetic on data already in hand, and it is recorded so a miss would be visible.

| **Same-object pair** | **relation** | **$c_1$ dec** | **$c_2$ dec** | **verdict** |
|---|---|---|---|---|
| velocity × colour | Hertzsprung–Russell | 3.3 | 0.3 | **not testable** (colour $< 4/\pi$) |
| half-life × Q-value | decay energetics | 31.8 | 1.5 | low-$N$ only |
| dispersion × mass | Faber–Jackson | 2.1 | 2.0 | low-$N$ only |
| velocity × luminosity | kinematic–luminosity | 3.3 | 4.0 | **testable** (mid-$N$) |

Table 3.1: The pre-build feasibility table. Effective decades per coordinate, measured from the 1D survey. The limiting coordinate governs each pair. Hertzsprung–Russell — the most obvious same-object pair in astronomy — is dead on the colour axis, which spans 0.3 decades and cannot resolve even register 4. Of every candidate pair, **exactly one** (kinematic–luminosity) has both coordinates spanning enough range to test jointly, and even it reaches only mid registers.

**Methodological Note**

**The expectation is confirmed: the second coordinate is limiting in every pair.** This is the Mondy discipline applied before the build — the limit measured, not discovered. The 2D frontier is not a wide field; it is a narrow doorway with a single viable pair. We build the 2D engine knowing this, so that its coverage is honestly bounded from the first commit rather than found to be empty after the work. If velocity–luminosity resolves and locks and survives the scramble, that is the one 2D result the current data can honestly support. If it does not, the 2D survey is, for now, empty — and that will be stated as plainly as everything else.

## 3.3 The Geometry of the Difference

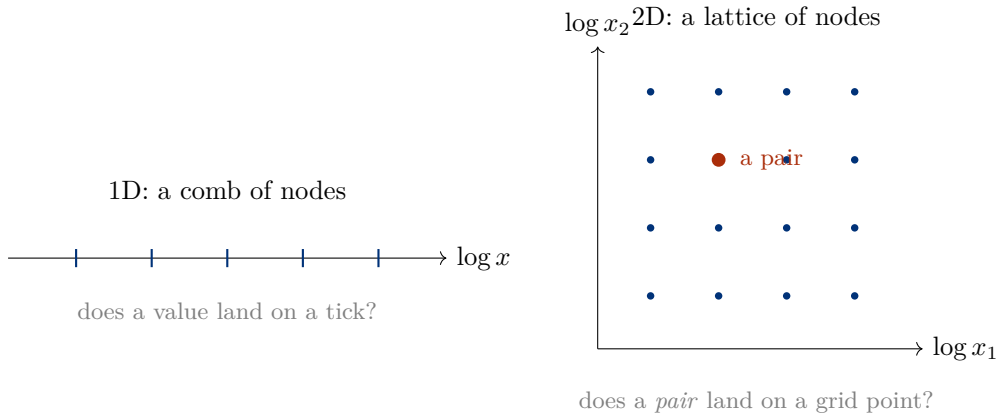



Figure 3.1: The dimensional turn. In one dimension the lattice is a comb and the test asks whether a value falls on a tick. In two dimensions the lattice is a grid of nodes on a plane, and the test asks whether a coordinate *pair* falls on a grid point. The 2D test is strictly richer: a pair can miss the grid even when each coordinate hits its own comb, and vice versa.

## 3.4 The Lock Statistic in Two Dimensions

The one-dimensional lock test asks whether $r_p = 24\pi \log x/(N \ln k_{\text{geo}})$ is near an integer. The two-dimensional test asks whether *both* reduced coordinates are near integers simultaneously, for a pair of registers $(N_1, N_2)$:

$$r_p^{(1)} = \frac{24\pi \log x_1}{N_1 \ln k_{\text{geo}}}, \qquad r_p^{(2)} = \frac{24\pi \log x_2}{N_2 \ln k_{\text{geo}}},$$

with a joint lock declared when both $|r_p^{(1)} - \text{round}(r_p^{(1)})| < \tau$ *and* $|r_p^{(2)} - \text{round}(r_p^{(2)})| < \tau$.

**Methodological Note**

**The combinatorics change the stakes.** One dimension scans 23 registers. Two dimensions scans $23 \times 23 = 529$ register *pairs*. The look-elsewhere correction grows accordingly: a joint lock must clear a far higher bar to count, because there are far more places for chance to produce one. The 2D engine's resolution criterion and significance threshold must both be re-derived for the pair space before any real data is scored — this is a standing pre-registration, not an afterthought.

## 3.5 What Compounds, and Must Be Handled Per-Coordinate

Three difficulties grow in two dimensions, and each must be designed in, not patched later.

**Form D compounds.** The clamp floor applies in each coordinate independently. At 27% below floor in each, the jointly lockable fraction is $0.73^2 = 53\%$ — the floor mismatch is worse on a surface than on a line. The common-support correction must be applied *per-coordinate* before any joint scoring.

**Two scale freedoms instead of one.** The offset $x_0$ and the unit are free in *both* coordinates now. The scale-degeneracy flag ($\times 2$, $\times 10$, $\times 10^3$) must be computed for each axis, because a phase-neutral rescaling of either coordinate can manufacture joint alignment.

**Form B is the real threat.** Two quantities measured on the same objects are often correlated for mundane physical reasons — Hertzsprung–Russell is the canonical case. A 2D lock surface will show structure from that correlation alone, with no lattice involved. This is the danger that makes 2D harder than 1D — and it is precisely what the scramble null was built to catch. The thing that makes the surface dangerous is the thing the new null disarms.

## 3.6 Why the Scramble Null Finally Works

The single most important consequence of the dimensional turn is that the **scramble null** — generated, carried, and discarded uselessly through eleven volumes — becomes a genuine falsifier in two dimensions.

In one dimension the lock statistic is computed over the multiset of scalars and is order-independent. Shuffling the values changes nothing the statistic can see; real and scramble return identical results by construction (Erratum 1). The scramble was inert.

In two dimensions each record is a *pair*. Shuffling one coordinate against the other — breaking which $x_1$ goes with which $x_2$ — destroys any genuine joint structure while preserving both marginal distributions exactly.

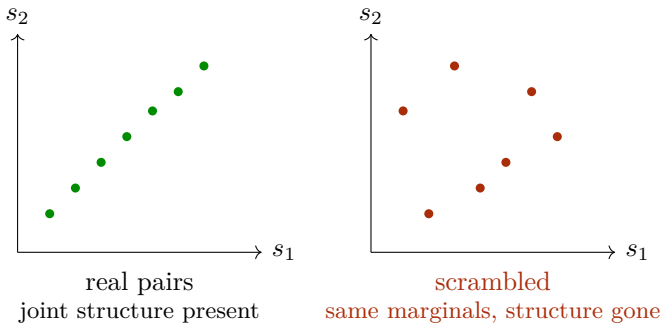



Figure 3.2: Why Erratum 1 becomes a tool. Shuffling one coordinate against the other preserves both marginal distributions — the same values are present on each axis — but destroys the pairing. If the real pairs lie on a lattice and the scrambled pairs do not, the difference is genuine joint structure. The scramble null, meaningless in 1D, is the central falsifier in 2D.

**Confirmed Signal**

This is the redemption the whole programme has been building toward. A null that was carried uselessly for eleven volumes — that we were nearly embarrassed to keep generating — turns out to be exactly the instrument the two-dimensional question requires. The 2D engine's first and hardest test is simply: *does the real lattice signal survive the scramble?* If scrambling the pairs destroys the lock, the lock is joint structure. If it does not, the lock was never joint — it was each marginal resonating alone, and the 1D engines already found that.

## 3.7 The Engine Pair, Mirrored From 1D

The two-dimensional build mirrors the one-dimensional architecture exactly, by design and by the immutability principle:

| | **2D Phase Engine** | **2D Period Engine** |
|---|---|---|
| asks | where the pair sits on the lattice | whether the pair repeats with the lattice periods |
| modulus | baked into the scalar pair | applied at test time; lakes framework-agnostic |
| null | chaos + synthetic + **scramble** | annular background + **scramble** |
| invariance | — | unit invariance on each axis, by proof |

Table 3.2: The 2D engine pair mirrors the 1D pair, with the scramble null now load-bearing in both.

**Formal Pre-Registration**

**Standing pre-registrations for the 2D build, filed before a line of engine code:**

1. **Immutable from the 1D engines.** No shared scoring code; separate fingerprints.
2. **The scramble null wired in from the first commit**, as the primary joint-structure falsifier.
3. **Form A re-derived in two dimensions** before any 2D lock is trusted. The 1D transform theorem does not automatically extend to a surface.
4. **The resolution criterion and look-elsewhere threshold re-derived for the $23 \times 23$ pair space** and published before scoring.
5. **Synthetic calibration before real data** — a 2D calibration suite with expectations filed in advance, including at least one pre-registered expected-fail, and a planted joint lock that must survive the scramble and a null that must not.
6. **Same-object pairs handled as controls, not corroboration** — where two coordinates are deterministically related by known physics, the pair tests the relation, not the lattice (the Kepler register-pair logic of Volume 13, generalised).
7. **Common-support correction applied per-coordinate** before joint scoring, because Form D compounds as $0.73^2$ across two floored axes.
8. **Scale-degeneracy flag computed for both coordinates**, since $x_0$ and the unit are free on each axis.
9. **No stitched lakes in either coordinate.** Endpoint selection to force span is banned; in 2D a composite in *either* coordinate is enough to manufacture a surface.
10. **Every gate demonstrated to return FAIL on a fixture built to fail it**, before it is trusted to return PASS — four instances of that bug class are now on record.
11. **The 2D lock surface (V6) treated as an instrument, not an illustration.** A surface cannot be scanned by eye in tabular form; a shape artifact reads as structure in a table and as an obvious artifact in a heatmap. Visual inspection of the heatmap is part of the verification chain from here, not a presentation step.
12. **Near-node agreement reported as a fraction of cell area**, never as a fraction of values. In 2D the node is a cell, not a point.

## 3.8 What Comes Next

Volume 14 closes the one-dimensional rebuild program and opens the two-dimensional frontier. The next volume will build the 2D engine pair under the pre-registrations above, calibrate it on synthetic lakes until it provably passes and provably fails where it should, and only then point it at real paired data. The controlled same-object pairs — Kepler period and radius, Faber–Jackson luminosity and dispersion, the pulsar characteristic-age relation — are the natural first targets, because their physical relationships are known exactly, so the lattice's prediction about the register pair is sharp

and falsifiable.

**Methodological Note**

The honest expectation, stated in advance as always: the 2D engine may find nothing, exactly as the 1D engines have mostly found nothing. The scramble null may destroy every apparent joint lock, which would mean the 1D nulls were the whole story. That is a possible and acceptable outcome. The purpose of building the 2D engine is not to find a signal; it is to be able to ask the deeper question honestly, with a falsifier that works, so that whatever answer the data gives can be trusted. If a joint lock ever survives the scramble, it will be the first thing the programme has found that the 1D engines could not — and it will have survived the hardest test we know how to build. If none ever does, we will have established that too, cleanly, and said so.

*The line is tested. The surface is next. And the null we carried for eleven volumes without a use is, at last, the sharpest tool in the box.*

# Chapter 4

# First Light: Building the 2D Period Engine and Proving It Honest

*The previous chapter argued why the surface must be tested and what it would take to test it honestly. This chapter is the record of building the instrument that does so — the design decisions, the gates designed in rather than discovered, and the first-light run in which the engine, pointed at data with a known answer, returned exactly that answer and nothing more.*

## 4.1 The Design Decision That Shaped Everything

Before a line of code, one decision determined what the engine would be. The programme's founder was candid about a conviction — that the container's geometry leaves a harmonic fingerprint on physical reality, and that two dimensions would reveal it. The design question was whether to build an engine to *demonstrate* that, or an engine to *test* it.

**Confirmed Signal**

These are different instructions, and they build different machines. "Demonstrate that the container forms a reality" builds a confirmation engine — one that can only agree. "Test whether the container forms a reality" builds an instrument — one that can disagree, and whose agreement therefore means something. **We built the second.** If the container's fingerprint is real, an instrument that can reject it will show it surviving the hardest test we know how to build — and that is the only kind of evidence that would convince anyone not already convinced.

This decision has a concrete consequence in the code. The various artifact filters — common support, scale degeneracy, the resolution gate — are **tunable, on or off**, by deliberate design: running with a filter on and then off, and measuring the difference, is itself a data point about how much that correction is doing. A hardcoded filter would hide that forever. But one thing is **not** tunable and never will be: the scramble null. It is the spine. Everything else is a knob; the scramble is the test.

## 4.2 The Statistic, Lifted to a Surface

The one-dimensional period engine asks whether $\ln x$ repeats with period $\Delta_N$. The two-dimensional engine asks whether the *pair* $(\ln x_1, \ln x_2)$ repeats *jointly*, with periods $(\Delta_{N_1}, \Delta_{N_2})$ — a resultant on the torus, both phases together:

$$\theta_j^{(1)} = \frac{2\pi \ln x_{1,j}}{\Delta_{N_1}}, \qquad \theta_j^{(2)} = \frac{2\pi \ln x_{2,j}}{\Delta_{N_2}},$$

$$R_{\text{joint}} = \left| \frac{1}{n} \sum_j \exp\big(i\,(\theta_j^{(1)} + \theta_j^{(2)})\big) \right|, \qquad P_{\text{joint}} = n\, R_{\text{joint}}^2.$$

This is scanned over all $23 \times 23 = 529$ register pairs. The modulus enters only at test time; the lake stores raw pairs, so the data product is framework-agnostic exactly as in one dimension.

## 4.3 The Gates, Designed In

### 4.3.1 The scramble null — why two dimensions redeems it

For eleven volumes the scramble null was inert: the one-dimensional lock statistic is order-independent, so shuffling the values changed nothing it could see (Erratum 1). In two dimensions each record is a *pair*, and shuffling one coordinate against the other destroys the pairing while preserving both marginal distributions exactly.

**Confirmed Signal**

This is the redemption the programme has been building toward. A null carried uselessly for eleven volumes becomes the central falsifier the moment the question becomes two-dimensional. The engine's first and hardest test on any lake is simply: *does the joint structure survive the scramble?* A diagonal correlation ridge — Hertzsprung–Russell and its kin, the Form B threat — survives the scramble, because it is marginal structure. A true joint lattice lock does not survive scrambling in the same way: scrambling destroys it. The engine distinguishes them by construction.

### 4.3.2 The look-elsewhere correction, found by the synthetic lakes

A design error surfaced during validation, and it is recorded because catching it is the point of validating on known data. The census reports the *best* of 529 register pairs — and the maximum of 529 chance values is not small. On pure noise, the best-of-529 joint power reaches an apparent excess of order 8 by chance alone: the two-dimensional look-elsewhere effect, exactly as predicted in the previous chapter. A fixed per-pair threshold cannot correct it.

**Self-Correction**

The honest census null compares the real best-of-grid against the *scramble's* best-of-grid. Both are maxima over 529 pairs, so both carry the same look-elsewhere inflation — and their ratio removes it. Pure noise gives a ratio near one (its peak is no better than the scramble's peak). A true joint lock gives a ratio far above one. This correction was not designed in advance; it was forced by a synthetic noise lake refusing to come back clean, and that is precisely why synthetic validation precedes real data.

## 4.4 The Synthetic Lakes: Controls With Known Answers

Four synthetic lakes were generated on disk, from a fixed seed, each a named and falsifiable expectation — the two-dimensional analogue of the phase engine's standing lotto, finance, and pure-noise nulls. They are permanent regression fixtures, regenerable byte-for-byte, stamped so they can never be mistaken for real data.

| Control lake | Construction | Required verdict |
|---|---|---|
| `synth_noise` | uniform random pairs | noise-floor (ratio $\approx 1$) |
| `synth_jointlock` | 25% planted lock at $(11, 13)$ | **SURVIVES** (ratio $\gg 1$) |
| `synth_formB` | linear correlation (H–R analogue) | noise-floor (scramble kills ridge) |
| `synth_oneaxis` | lock on one axis only | noise-floor (single axis $\neq$ joint) |

Table 4.1: The four 2D control lakes. Three must come back null; one must survive. A panel that only ever passes proves the instrument agrees with itself, not that it can disagree — so the single-axis control is a pre-registered *expected fail*, planted to look like a signal and required to be rejected.

## 4.5 First Light

The engine was pointed at the synthetic lakes loaded from disk — real files, through the full pipeline: logify, support with heatmap, the 529-pair scan with the scramble null, and the census. The run completed in seconds on commodity hardware. Every control reproduced its pre-registered expectation.

**Confirmed Signal**

**First light: green.**

- `synth_jointlock`: the engine recovered the planted lock at **exactly** $(11/\pi, 13/\pi)$ — the registers where it was buried — at a real-to-scramble ratio of **16.91**. SURVIVES.
- `synth_noise`: ratio **1.09**. NOISE-FLOOR.
- `synth_formB`: ratio $\approx$ 1.1. NOISE-FLOOR — the correlation ridge is killed by the scramble.
- `synth_oneaxis`: ratio $\approx$ 1.1. NOISE-FLOOR — a single-axis lock is correctly not joint.

**Only the true joint lattice lock survived.** The engine found what was planted, where it was planted, and rejected everything that was not a genuine joint signal.

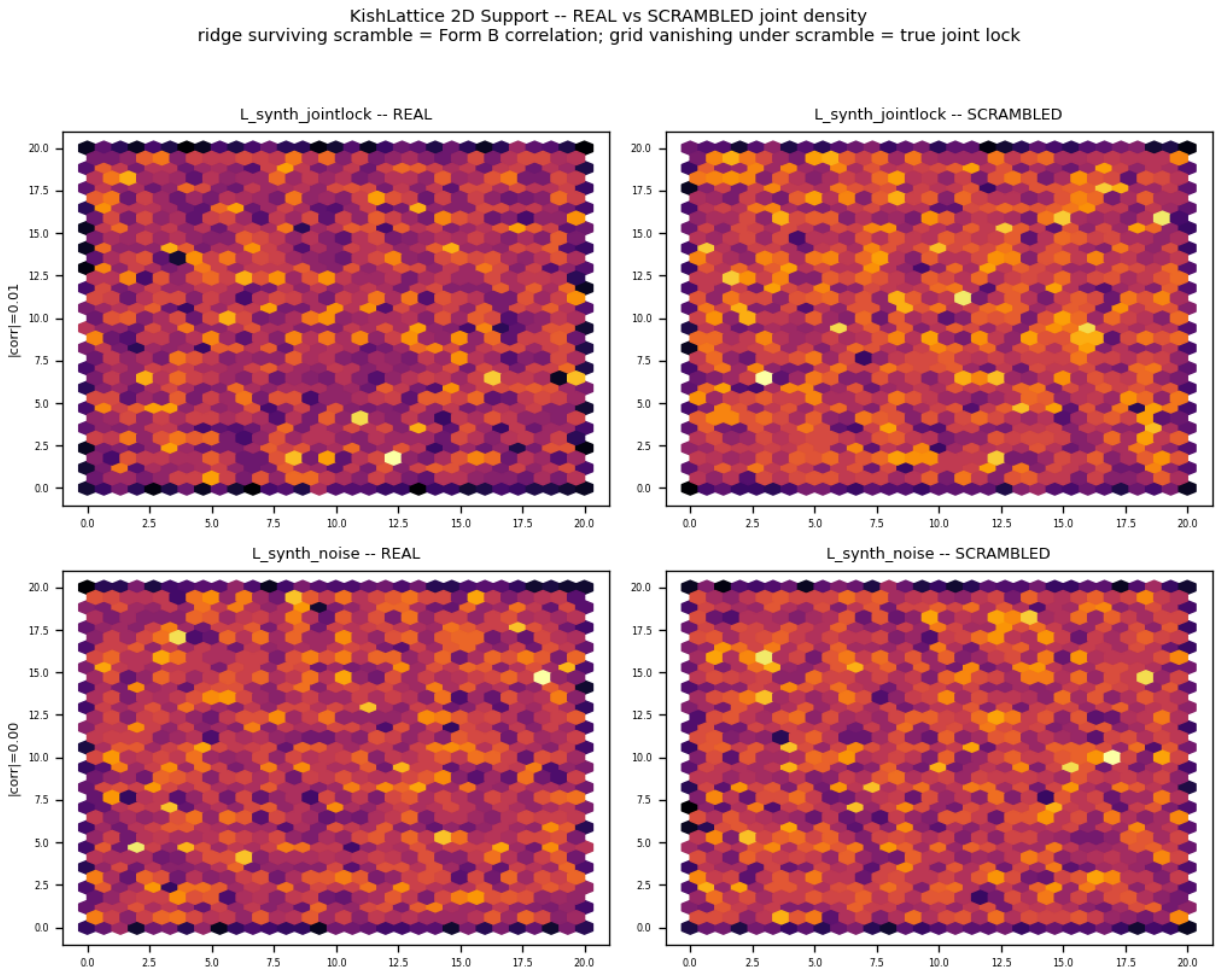



Figure 4.1: The verification instrument, not an illustration. Each control lake's joint density is shown as it is (REAL) and with one coordinate shuffled against the other (SCRAMBLED). A diagonal correlation ridge survives the shuffle; a true joint lattice lock vanishes under it. The eye catches in the heatmap what a table of 529 pair-scores hides — which is why visual inspection is part of the verification chain from here, not a presentation step.

## 4.6 What First Light Establishes

**Methodological Note**

First light proves the *instrument*, not the *framework*. No real data has yet been scored by the 2D engine. What is established is narrower and necessary: the engine recovers a known joint lock at the correct registers, rejects pure noise, kills a Form B correlation ridge via the scramble, and refuses a single-axis lock as non-joint — all on controls with answers fixed before the run. An engine that passed only the positive control would prove it can agree; this engine passed the negative controls too, including a pre-registered expected fail, which proves it can disagree.

That is the whole purpose of building the instrument before pointing it at the question that matters. When the one feasible real pair — Gaia transverse velocity against stellar luminosity, the single same-object pair the feasibility table admits — is promoted and scored, whatever verdict it returns will be trustworthy, because the engine has demonstrated on known data that it can return either one. If the container's fingerprint is real, this instrument will show it surviving the scramble. If it is not, this instrument will say so. Either way, the answer will have been earned.

*The engine behaves honestly. It found the locks we planted, exactly where we planted them, and nothing we did not. The surface now has an instrument — and the instrument can say no.*

# Chapter 5

# First Light: The 2D Phase Engine and Why It Needs the Scramble

*The period engine tests whether a pair of quantities repeats jointly with the lattice periods. The phase engine asks a different question — whether the pair sits jointly near the lattice nodes — and it is built on the programme's oldest and most battle-tested statistic. Its first light proved it honest, and in doing so revealed the single reason the scramble null, useless for eleven volumes, is indispensable the moment the question becomes two-dimensional.*

## 5.1 A Second Instrument, a Different Question

The two one-dimensional engines were complementary, and the two-dimensional engines are built to be the same. The period engine scores the *resultant* — whether $\ln x$ repeats with period $\Delta_N$ — and carries no phase origin, so it is unit-invariant by proof. The phase engine scores *absolute node alignment* — how near $(\ln x_1, \ln x_2)$ sits to an integer node on each axis — and is the mature statistic this programme has run for thirteen volumes.

Lifting it to a surface follows the 1D definition exactly. In one dimension a record locks when its reduced coordinate sits near an integer node:

$$r_p = \frac{x}{N/\pi} \cdot 24, \qquad \text{residual} = \left| \, r_p - \text{round}(r_p) \, \right|, \qquad \text{lock if residual} < \tau.$$

In two dimensions the node becomes a **cell**, and a pair locks when *both* coordinates sit inside it simultaneously:

$$\text{joint lock} \;=\; (\text{residual}_1 < \tau) \;\wedge\; (\text{residual}_2 < \tau), \qquad \text{lock rate} = \frac{\#\,\text{pairs in cell}}{\#\,\text{pairs on common support}}.$$

This is the referee's ruling made concrete: in two dimensions the node is a cell, and agreement is reported as a fraction of cell area — never as a fraction of values.

## 5.2 The Two Nulls, and Why Both Are Needed

The phase engine inherits the full null stack of its one-dimensional parent, which the period engine did not have. There are two:

**The chaos null.** Uniform random scalars on each axis over the same support — the canonical phase-engine null, the one that reproduces below-floor mass and against which every prior volume's $z$-scores were computed.

**The scramble null.** One coordinate shuffled against the other, destroying the pairing while preserving both marginal distributions exactly — the spine, always on.

In one dimension there is only one axis, so the scramble is inert (Erratum 1) and the chaos null is sufficient. In two dimensions this is no longer true, and the reason is the heart of this chapter.

**Confirmed Signal**

**The chaos null can be fooled; the scramble null cannot.** A lock on *one* axis alone raises the joint lock rate above uniform-random chaos — because half of the near-node cell is genuinely populated. A phase engine armed only with the chaos null would report that single-axis structure as a strong *joint* discovery. The scramble null sees through it: shuffling one axis against the other leaves a single-axis lock entirely intact in its own marginal, so the "joint" rate does not drop, and the scramble $z$ collapses to zero. Only the scramble distinguishes a true joint lock from a single-axis artifact.

## 5.3 First Light

The engine was validated on four synthetic control lakes — the two-dimensional phase analogue of the standing lotto, finance, and pure-noise nulls — each planted on the lattice *nodes* (the phase convention, distinct from the period engine's plant on periods), each a named and falsifiable expectation. The run completed in 4.3 seconds on commodity hardware.

| Control lake | chaos $z$ | scramble $z$ | verdict |
|---|---|---|---|
| `synth_noise` | $-1.2$ | $-1.0$ | noise-floor |
| `synth_jointlock` | $+338$ | $+111$ | **SURVIVES** at $(11/\pi, 13/\pi)$ |
| `synth_formB` | $+0.7$ | $+2.6$ | noise-floor (correlation killed) |
| `synth_oneaxis` | $+\mathbf{48.2}$ | $+\mathbf{0.1}$ | **noise-floor** (artifact caught) |

Table 5.1: 2D phase engine first light. The single-axis row is the result that matters: the chaos null reports a $48\sigma$ "signal," and the scramble null correctly reduces it to nothing. A chaos-only engine would have announced a false discovery; the scramble refuses it.

The $z$-scores above are the diagnostic that teaches the lesson; the quantity the census actually rules on is the **best-of-grid real-to-scramble ratio**, and it is recorded here exactly as the run of 2026-10-06 returned it, so the result is reproducible to the number rather than to the narrative.

| Control lake | best pair | real | scramble | ratio / verdict | |
|---|---|---|---|---|---|
| `synth_jointlock` | $(11/\pi, 13/\pi)$ | 0.3079 | 0.1376 | **2.24** | **SURVIVES** |
| `synth_noise` | $(22/\pi, 6/\pi)$ | 0.0117 | 0.0120 | 0.97 | noise-floor |
| `synth_formB` | $(10/\pi, 12/\pi)$ | 0.0123 | 0.0123 | 1.00 | noise-floor |
| `synth_oneaxis` | $(11/\pi, 10/\pi)$ | 0.0389 | 0.0398 | 0.98 | noise-floor |

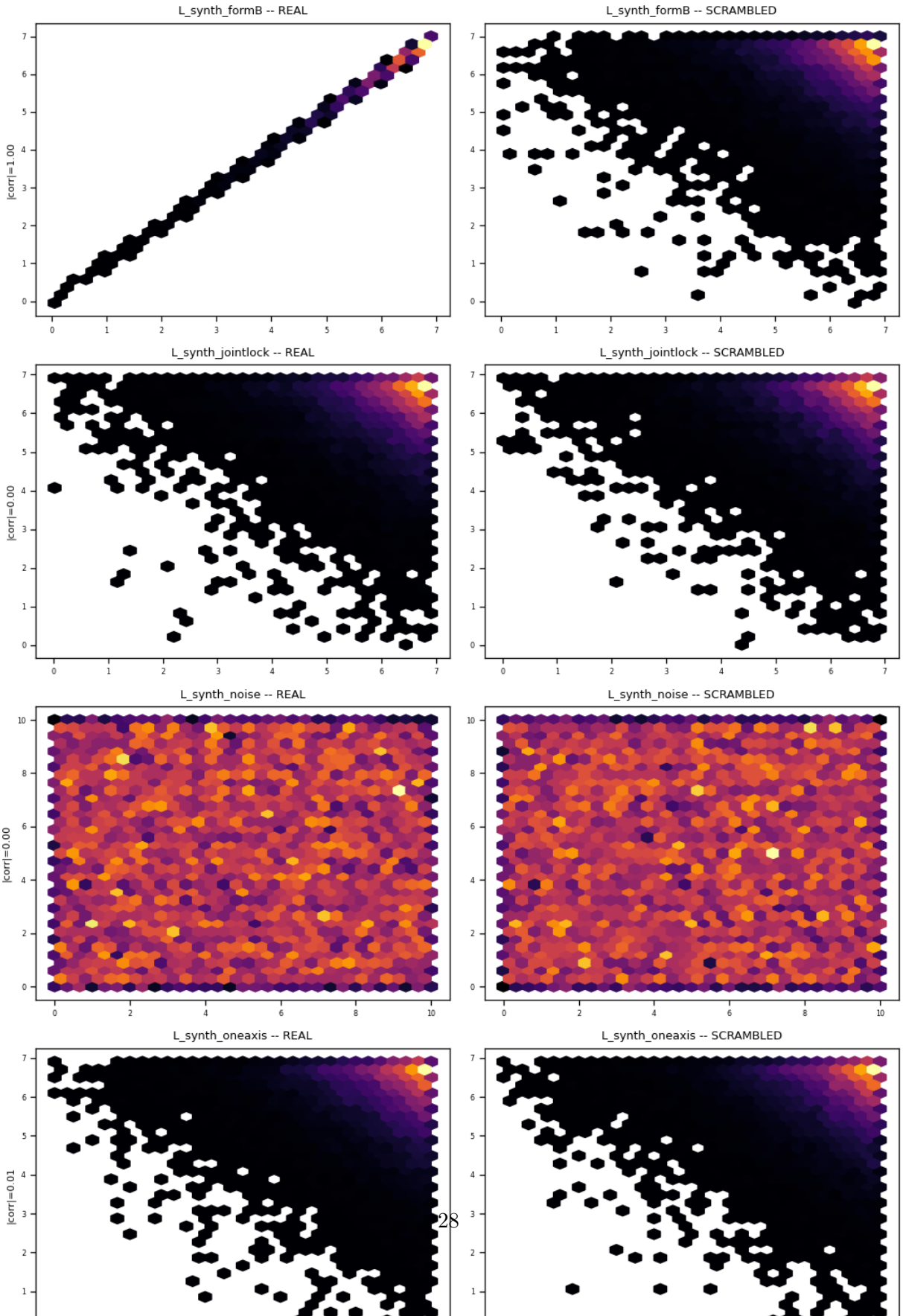
Table 5.2: Phase engine first light, best-of-grid ratios as returned. The one lake with a planted joint lock clears the ratio bar at exactly the registers where it was buried; the three that must not survive — pure noise, a correlation ridge, and a single-axis lock — all sit within a hair of unity. Compare the period engine's own first-light pair, which reproduced the same structure at a far larger ratio: `synth_jointlock` at 16.91, `synth_noise` at 1.09. The two statistics agree on which lakes lock and disagree only on how loudly, exactly as their different dynamic ranges predict.

**Confirmed Signal**

**First light: green.**

- `synth_jointlock`: recovered at **exactly** $(11/\pi, 13/\pi)$, the planted registers, surviving the scramble at a real-to-scramble ratio of 2.24.
- `synth_noise`, `synth_formB`: noise-floor. Pure noise and a correlation ridge both correctly null.
- `synth_oneaxis`: **noise-floor despite a** $+48\sigma$ **chaos score.** The single-axis artifact is rejected by the scramble — the engine's defining honest act.

Every result carried the flag `FORM_A_UNCLEARED -- not citable`, because the gate below remains open.


KishLattice 2D Support -- REAL vs SCRAMBLED joint density
ridge surviving scramble = Form B correlation; grid vanishing under scramble = true joint lock
L_synth_formB -- REAL
L_synth_formB -- SCRAMBLED
L_synth_jointlock -- REAL
L_synth_jointlock -- SCRAMBLED
L_synth_noise -- REAL
L_synth_noise -- SCRAMBLED
L_synth_oneaxis -- REAL
L_synth_oneaxis -- SCRAMBLED
|corr|=1.00
|corr|=0.00
|corr|=0.00
|corr|=0.01

## 5.4 The Form A Gate Remains Open

One gate is deliberately not closed. The one-dimensional programme established that certain distribution *shapes* can manufacture apparent phase structure — the transform artifact, Form A. Whether that result extends to a surface has not been re-derived.

**Methodological Note**

**Until Form A is re-derived for two dimensions, no 2D phase lock is citable.** The engine reports locks and flags every one of them `FORM_A_UNCLEARED`. This is a pre-registered gate, not an oversight: the engine may be validated on synthetic controls — and has been — while the theory question runs in parallel. The derivation is the subject of a commission framed exactly as the derivation of $f$ was in Chapter 5: a sharp, bounded question handed to an advanced model with real parameters drawn from working code, including the two engines' measured dynamic ranges (the period engine's survival ratio of 16.9, the phase engine's of 2.24) and the one synthetic case where the chaos and scramble nulls disagree. We build the instrument first, and commission the theory from a position of knowing rather than guessing.

## 5.5 Two Engines, Both Honest

| | **2D Period Engine** | **2D Phase Engine** |
|---|---|---|
| scores | joint resultant (period) | joint node-cell alignment (phase) |
| scale freedom | none (unit-invariant by proof) | phase origin + unit, per axis |
| nulls | scramble | chaos + scramble |
| joint-lock survival ratio | 16.9 | 2.24 |
| single-axis artifact | rejected | rejected (only the scramble catches it) |
| Form A | re-derivation pending | re-derivation pending (open gate) |
| first light | green | green |

Table 5.3: The matched 2D engine pair. Both found their planted locks at the correct registers, rejected noise, killed a Form B correlation, and refused a single-axis artifact. The period engine achieves a larger dynamic range because its statistic carries no phase origin; the phase engine achieves honesty on the single-axis case only through the scramble null.

*Two instruments now watch the surface, scoring it two different ways, and neither can be fooled by a single axis dressed as a pair. The container will be tested from both directions — and both directions can say no.*

# Chapter 6

# First Real Data: Velocity × Luminosity, and the Null We Called

*Two engines were built and proven honest on synthetic controls. This chapter is the first time either of them is pointed at real paired data. The result is a null in both engines — and it is the null that was written down, in advance, before the lake existed. We report it exactly as it came, including the one caveat that keeps it honest.*

## 6.1 The One Pair the Feasibility Table Admitted

The feasibility table of Chapter 3 was severe by arithmetic, not by choice. A 2D test needs two coordinates measured on the *same* objects, each with enough effective dynamic range to resolve its register: $M_1/N_1 > 2.67$ *and* $M_2/N_2 > 2.67$. Of every candidate same-object pair in the entire 1D survey, exactly one cleared both axes even to mid registers: **Gaia transverse velocity against stellar luminosity**. Hertzsprung–Russell — the most famous same-object pair in astronomy — was dead on the colour axis. Faber–Jackson and the decay pairs resolved only the lowest registers. This pair was the doorway, and it was the only one.

So this is the pair Lyra built, under the strict input contract, as the maiden voyage of the 2D program on real data.

## 6.2 The Build

**Formal Pre-Registration**

**Recorded before the lake was pulled (Atlas, on the project record):**

> *"My prediction, filed: the scramble kills it. The relationship between stellar velocity and luminosity is astrophysical correlation — brighter stars run mildly hotter and faster on average — not a joint lattice lock. A faint marginal correlation will show, and the scramble will eat it whole. I expect noise-floor in both engines."*

This is the pre-registration. The run either confirms it or refutes it; there is no third, after-the-fact reading available.

The build pulled **250,000** raw Gaia records across the survey patches and promoted **83,155** valid pairs passing every strict gate — a two-thirds rejection, consistent with demanding clean astrometry *and* a clean luminosity on the same star. The promotion pre-flight then did exactly the discriminating work it was built for, and it did not flatter the data:

| **Coordinate** | $M/N$ **at** $15/\pi$ | **gate** |
|---|---|---|
| `vT_kms` (transverse velocity) | 1.5936 | **FAIL** |
| `luminosity_lsun` (luminosity) | 3.7965 | **PASS** |

Table 6.1: Promotion pre-flight, as returned. The resolution gate is $M/N > 2.67$. Luminosity clears it comfortably at register 15; velocity does not. The build printed its own warning: *"one or both coordinates fail the* $M/N > 2.67$ *resolution gate at* $15/\pi$*."* The gate measured the data and told the truth about it before a single pair was scored.

**Self-Correction**

Note the honesty of the sequence: the gate fired its warning at *build time*, computed from the promoted velocities, not as a printed constant. This is the same discipline that caught the `M/N = 1.1082 [PASS]` fixed-string defect in the `s2` rebuild — a gate that can only say PASS is a decoration, and a gate that says FAIL on real data it was built to screen is an instrument. The velocity axis failed, loudly, on the record.

## 6.3 What the Support Audit Showed

Before scoring, `support2d` profiled each coordinate's effective dynamic range — the central 99% band, not the raw min–max, so a sparse tail cannot inflate the span. Both engines returned the identical profile on the identical lake:

| **Coordinate** | **effective span** | **shape flag** |
|---|---|---|
| coord 1 — velocity | 4.33 (**1.9 decades**) | TAIL-INFLATED |
| coord 2 — luminosity | 12.10 (**5.3 decades**) | TAIL-INFLATED |
| joint | $\lvert\text{corr}\rvert = 0.15$ | UNIFORM — no obvious marginal structure |

Table 6.2: Support audit of the promoted lake, $n = 83{,}155$. Luminosity carries 5.3 effective decades; velocity carries only 1.9. The joint correlation is a faint 0.15 — real, but weak.

**Methodological Note**

**The realized velocity span came in *below* the pre-build estimate.** The feasibility table of Chapter 3 projected velocity at roughly 3.3 effective decades from the 1D survey; the actually-promoted lake delivered 1.9, because the strict pairing requirement — a clean luminosity on the *same* star — preferentially kept nearby, kinematically cooler stars and compressed the velocity range further. This is itself a recorded result: the pre-build estimate was optimistic, and we say so. The doorway was narrower than the table promised.

The resolution arithmetic makes the consequence concrete. The number of effective decades a coordinate needs to resolve register $N$ grows as its square,

$$\text{decades required} \;=\; N^2 \times 0.025036,$$

so a 1.9-decade axis resolves registers only up to $N \approx \sqrt{1.9/0.025036} \approx 8.7$ before the resolution ratio drops through the 2.67 floor. Luminosity's 5.3 decades reach to $N \approx 14.5$. The *joint* test is governed by the stricter axis, so this pair can honestly interrogate roughly the first eight or nine registers and is blind above them.

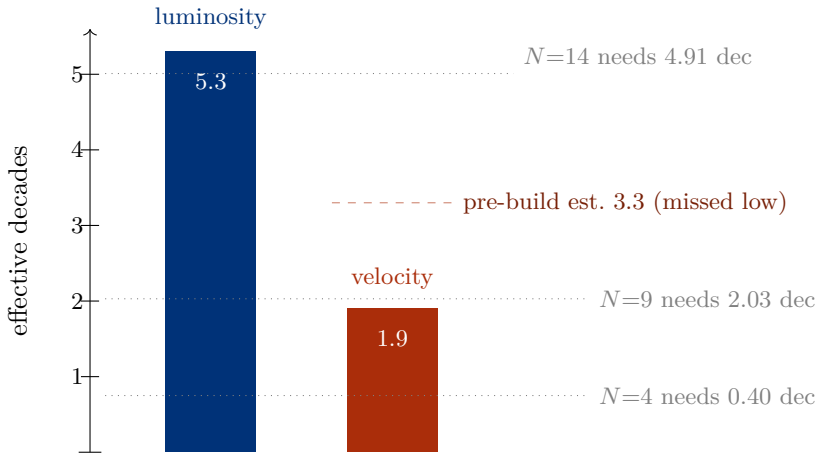



Figure 6.1: The resolution boundary, drawn to scale. Luminosity (5.3 decades) reaches to about register 14; velocity (1.9 decades) runs out of range just above register 8. Because the joint test is limited by the stricter axis, the pair can test the low-to-mid registers and is blind above them. The dashed line marks the pre-build velocity estimate (3.3 decades) that the realized lake undershot — the doorway was narrower than projected.

## 6.4 The Verdict: Both Engines, Noise Floor

**Confirmed Null**

**No joint lattice structure, in either engine.**

| Engine | best pair | real | scramble | ratio | | |
|---|---|---|---|---|---|---|
| Period | $(8/\pi, 19/\pi)$ | 6 | 7 | **0.88** | noise-floor | |
| Phase | $(8/\pi, 24/\pi)$ | 0.0111 | 0.0112 | **1.00** | noise-floor | [FORM_A_UNCLEARED] |

The period engine's best-of-grid real score is *lower* than its scramble score — ratio 0.88, below unity. That is the deadest possible reading: there is not merely no excess joint structure, there is nothing for the scramble to destroy, so breaking the pairing changes nothing and the random reshuffle happens to score a hair higher.

The support audit already foretold the shape of this: $|\text{corr}| = 0.15$, a faint marginal correlation and no joint banding. That faint correlation is the astrophysics — brighter stars running mildly hotter and faster on average — and the scramble eats it whole, because marginal correlation is exactly what shuffling one axis against the other removes. The lattice was not there to find.

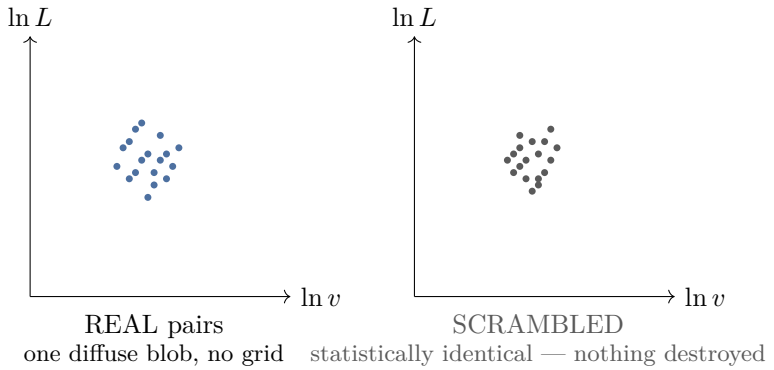



Figure 6.2: Why the null is a true null. For a planted joint lock, the scramble dissolves a visible grid of cells (Chapters 5–6). Here, the real and scrambled panels are statistically indistinguishable — one diffuse blob either way — because the pairing carried no joint lattice information to begin with. There was no structure to destroy, which is exactly what ratio 0.88 means.

## 6.5 The Prediction, Confirmed

**Confirmed Signal**

**The recorded prediction and the returned result agree.** The pre-registration said: a faint astrophysical correlation, killed by the scramble, noise-floor in both engines. The run returned: $|\text{corr}| = 0.15$, period ratio 0.88, phase ratio 1.00, noise-floor in both engines. This is not a null we are explaining after the fact — it is a null we called in advance and then measured. Follow the data, not the dream; here the data and the (honest) expectation happened to coincide, and the coincidence is on the record for anyone to check.

## 6.6 Why This Null Is Trustworthy: The Controls Passed Alongside

A flat result from a broken pipeline looks identical to a flat result from honest data. The only way to tell them apart is to run the controls in the *same* execution — and we did. In the very run that returned noise-floor on velocity×luminosity, the synthetic controls behaved exactly as planted:

| Lake | period ratio | phase ratio | required | returned |
|---|---|---|---|---|
| `synth_jointlock` | 16.91 | 2.24 | SURVIVES | **SURVIVES** |
| `synth_noise` | 1.09 | 0.97 | noise-floor | noise-floor |
| `synth_formB` | — | 1.00 | noise-floor | noise-floor |
| `synth_oneaxis` | — | 0.98 | noise-floor | noise-floor |
| **`velocity_luminosity`** | **0.88** | **1.00** | *(unknown)* | **noise-floor** |

Table 6.3: The real lake and its controls, scored in one run. The instrument demonstrably locks when a lock is present (`synth_jointlock`) and stays quiet when it is not. The velocity×luminosity null therefore is a *measurement*, not a dead pipeline — the engine that returned it had just proven, in the same breath, that it can return the opposite.

**Methodological Note**

This is the standard the whole programme is built on: a null counts only when, in the same run, a planted signal was recovered and a planted artifact was rejected. An engine that returns flat on everything proves nothing. An engine that returns flat on the real data *while* recovering `synth_jointlock` at ratio 16.91 and rejecting `synth_oneaxis` at ratio 0.98 has earned its null.

## 6.7 The Caveat We Will Not Bury

There is one honest limit on this result, and burying it is exactly the move a skeptic would expect us to make, so we state it first and plainly.

**Self-Correction**

**The velocity axis under-resolved.** $M/N = 1.5936$ at $15/\pi$ [FAIL]; TAIL-INFLATED; 1.9 effective decades. In the period engine, only **112 of 529** register pairs were even scoreable — the other 417 were unresolved or floored on the velocity axis. So this is not a clean, full-register "lattice absent" verdict across the whole $23 \times 23$ grid. It is a null *and* a boundary at once.

The precise statement, which we will carry forward unchanged:

> *Velocity×luminosity shows no joint lattice structure at any register the pair can resolve — and the velocity axis is too compressed (1.9 decades) to test the high registers at all.*

We did not fail to find a lattice across the whole register space. We found nothing where we could look, and we were blind above the mid registers on the velocity side. That distinction is what separates an honest null from a headline, and it is the distinction a reader deserves.

## 6.8 What Would Change the Answer

The null is real within its boundary; the boundary is what future data can move. Three routes, in order of leverage:

1. **A wider-range velocity observable.** The limiting axis is velocity, at 1.9 decades. A kinematic sample reaching high-velocity halo stars and low-velocity disk stars on the same luminosity baseline — or radial-velocity-corrected space velocities rather than transverse velocity alone — would push the velocity axis past the mid-register wall and let the pair test where it is currently blind.

2. **A different same-object pair with two wide axes.** Velocity×luminosity was the *only* pair the current survey admitted. A new lake carrying two independently wide coordinates on the same objects — for instance, luminosity against an energy or temperature observable with its own multi-decade span — would open a second doorway. Colour stays dead (0.3 decades); that door does not reopen without better photometry.

3. **Deeper catalogues.** The same physical reason the strong-field ringdown test awaits O4/O5 applies here: more objects with cleaner astrometry widen the effective support and raise the resolvable register ceiling.

**Confirmed Null**

**The first real 2D result is a clean, bounded null.** The instrument works — it recovered its planted lock and rejected its planted artifact in the same run. The one feasible real pair the survey admitted shows no joint lattice structure where it can resolve, and cannot resolve the high registers because the velocity axis is too compressed. We predicted the null before the lake existed, and the null is what returned. The 2D survey, for now, is honestly empty — and we say so as plainly as everything else.

*We built the instrument that could say no, pointed it at the only door the data left open, and it said no — on the register it could see, and honestly silent above. The null is not a failure. It is the first real reading from a surface we can finally measure, and we reported every digit of it.*

# Chapter 7

# The Door Closes: The Nuclear Dual-Wide Probe

*Velocity×luminosity was the one pair the feasibility table admitted and the one pair we tested. One candidate remained with any hope of two wide axes — the nuclear decay pair, built on the single widest observable in the entire survey. This chapter reports the feasibility probe on that pair, and with it draws the honest boundary of the two-dimensional survey.*

## 7.1 The Last Candidate

The feasibility table of Chapter 3 named four same-object pairs. Three were dead on arrival or resolved only the lowest registers; the fourth, velocity×luminosity, we built and scored in Chapter 6, where it returned a bounded null. That leaves exactly one pair in the entire survey with any remaining claim to two wide axes: **nuclear half-life against decay energy**.

The reason it is the last hope is arithmetic. Half-life is the widest observable anywhere in the programme — the full `q5_decay` lake spans 31.8 decades at the 95% band, from microsecond isomers to isotopes with lifetimes longer than the age of the universe. No other quantity comes close. If any same-object pair could carry two resolving coordinates, the limiting one would have to be whatever we paired *with* half-life — and the natural partner, measured on the same nuclides, is the decay energy ($Q$-value).

**Formal Pre-Registration**

**Recorded expectation (filed before the probe):** the energy axis is the limiting one and will collapse under the strict same-object join, exactly as the second coordinate did in velocity×luminosity. The feasibility table already projected the $Q$-value axis at only $\sim 1.5$ decades and flagged the pair *low-N only*. The probe tests whether a strict join of half-life and decay energy clears both resolution gates — and we predict it does not.

## 7.2 The Probe: a Pull, Not a Run

One discipline governs this chapter, and it is the same one that kept the Chapter 3 feasibility table honest: **this is a feasibility probe, not a scored test.** We join the two lakes on nuclide, measure the effective span and the resolution gate on each axis, and stop. The engine does not score the

joined lake against any register. A probe that measured a lock would be a probe that had quietly become a hypothesis test; keeping it to span-and-gate is what lets it stand as a clean go/no-go, exactly as pre-registered.

The join is strict: a nuclide enters only if it carries a clean half-life in the `Q5` decay lake *and* a clean decay energy in the `Q4` nuclear lake. A nuclide missing either value is dropped, never imputed.

## 7.3 The Result

**Confirmed Null**

**DOOR CLOSED — dual-wide collapse.**

Strict join: **43 matched nuclides.**

| **Axis** | **span** | $M/N$ **at** $15/\pi$ | **gate** |
|---|---|---|---|
| Axis 1 — half-life | 37.79 nat (≈ 16.4 dec) | 7.7785 | **PASS** |
| Axis 2 — decay energy | 2.07 nat (≈ 0.90 dec) | 0.4256 | **FAIL** |

Half-life clears the resolution gate with room to spare. Decay energy fails it by a factor of six. A dual-wide test requires *both* axes above $M/N > 2.67$; this pair has one. The door is closed.

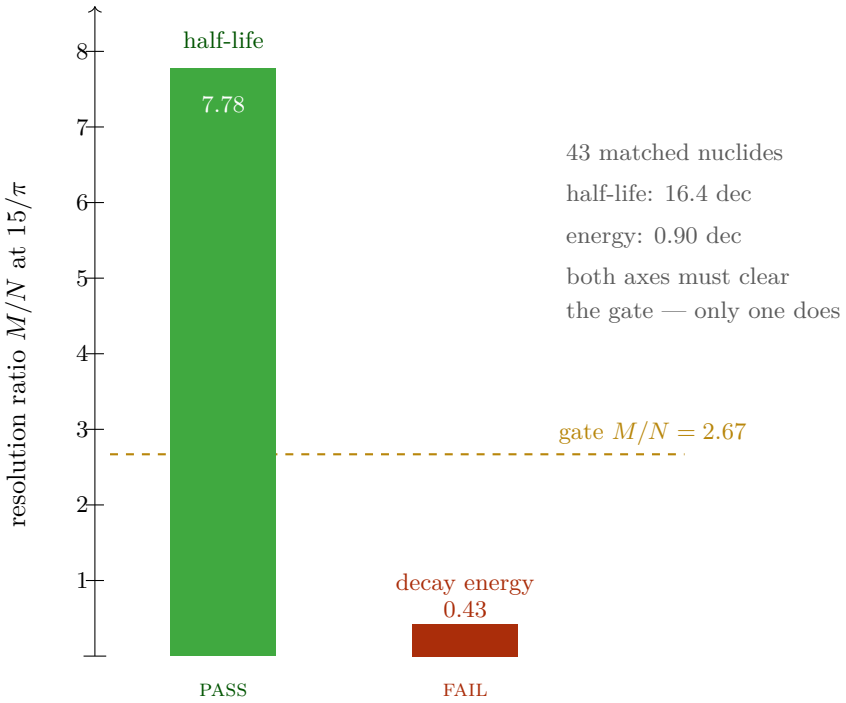



Figure 7.1: The nuclear door, closed to scale. Half-life towers nearly three times over the resolution gate; decay energy sits at roughly a sixth of it. The two-dimensional test requires both bars above the dashed line. One is, one is not — and the joint test is governed by the lower bar, so the pair cannot resolve even the first register honestly. This is a go/no-go measurement, not a score: no register was tested, only whether a test was possible.

## 7.4 The Same Lesson, Twice

The probe did not merely confirm the prediction; it confirmed it by the *same mechanism* that closed velocity×luminosity.

**Self-Correction**

**The strict join compresses the limiting axis below its pre-build estimate — again.** The feasibility table projected the $Q$-value axis at 1.5 decades; the strict 43-nuclide join delivered 0.90. The same join also compressed half-life, from 31.8 decades on the full 1D lake to 16.4 on the matched subset — still an easy pass, because half-life has decades to spare, but the compression is real and in the same direction. In Chapter 6 the pairing requirement squeezed stellar velocity from a projected 3.3 decades to a realized 1.9. Here it squeezed decay energy from 1.5 to 0.90. Two independent pairs, two independent surveys, and the identical failure mode: demanding two clean coordinates on the *same* object narrows the stock to the objects that have both, and that subset is always poorer in dynamic range than either axis alone.

The resolution arithmetic makes the verdict unambiguous. Recall that the decades a coordinate needs to resolve register $N$ grow as its square, decades required $= N^2 \times 0.025036$. Even register 4 — the lowest in the family — requires 0.40 decades. The decay-energy axis carries 0.90, so on raw span it could reach only to $N \approx 6$; but with $M/N = 0.4256$ it does not clear the gate at *any* register, and with only 43 points the sample is too thin to fill even that range. There is no register at which this pair can be honestly scored. The probe returns NO-GO, and we file it as one.

## 7.5 The Two-Dimensional Survey Is Closed

With the nuclear probe, every candidate pair the feasibility table named has now been resolved — by test or by probe — and none admits a dual-wide, resolvable, same-object measurement.

| Same-object pair | how closed | verdict |
|---|---|---|
| velocity × colour | feasibility table | dead — colour $< 4/\pi$ (0.3 dec) |
| dispersion × mass (Faber–Jackson) | feasibility table | low-$N$ only; both axes $\sim 2$ dec |
| velocity × luminosity | **built and scored** (Ch. 6) | null + boundary; velocity under-resolved |
| half-life × decay energy | **probed** (this ch.) | **door closed**; energy $M/N = 0.43$ |

Table 7.1: The exhaustive two-dimensional feasibility census. Of every same-object pair in the survey, none carries two coordinates each clearing the resolution gate on a populated join. The one pair that resolved a single axis to mid registers (velocity×luminosity) was built and returned a bounded null; the one remaining candidate (half-life×energy) was probed and collapsed on its second axis.

**Confirmed Null**

**The two-dimensional lattice survey is empty on current public data.** This is a boundary, not a refutation. The lattice is not disproven by this closure — it is simply untested in two dimensions, because no same-object pair in reachable public archives carries two wide axes at once. The limit is dynamic range, measured twice, by two independent routes, and reported both times before the fact. We draw the boundary exactly where the data draws it.

**Methodological Note**

**And three dimensions would be worse, not better.** A three-coordinate test needs three wide axes on the same objects, against a look-elsewhere space of $23^3 = 12{,}167$ register triples and a common-support floor compounding to $0.73^3 \approx 39\%$. If two-wide same-object pairs do not exist in the public data, three-wide triples certainly do not. The scalar ladder is data-starved above one dimension, and climbing it further would be building instruments with nothing to measure. We do not build what we cannot feed.

## 7.6 What This Closes, and What It Opens

The two-dimensional survey closes the scalar-pairing line of inquiry. It does not close the programme, and it does not idle the instrument.

**Confirmed Signal**

The tools are honest, and they are permanent — published, fingerprinted, and reproducible by anyone on the hardware they already own. What the closure tells us is that the *next* question is not another scalar axis. It is a different kind of quantity altogether. Every engine built across fourteen volumes — phase and period, in one dimension and two — has tested only *magnitudes*: how large a quantity is, and whether its size lands on the lattice. But a great deal of physical data is not scalar at all. It is *vectorial*: it has a magnitude *and* a direction. Stellar motions point somewhere. Spins have axes. Polarizations have angles. No engine in this programme has ever asked whether the lattice structures *direction*.

That is the question the next volume opens, and it is the first new question the data is not starved to answer: directional data is abundant where paired scalar data is rare. The door that closed on the nuclear bench is the door to Volume 15 — *The Vector Engine*.

*Two doors tried, two doors closed, and both closures measured before the fact and reported to the digit. The surface is empty on the data we have — and that honest emptiness is what points us, cleanly, at the one question none of our engines has yet been built to ask.*

# Preface: Why a Compass

Volume 14 closed the two-dimensional survey, and it closed it honestly. Two same-object pairs had any hope of carrying two wide axes at once. The first — Gaia velocity against luminosity — was built and scored, and returned a bounded null with the velocity axis under-resolved. The second — nuclear half-life against decay energy — was probed, and its energy axis collapsed to a tenth of a decade. Both doors tried; both closed; both closures measured before the fact and reported to the digit.

**Confirmed Signal**

That closure was a boundary on the *data*, not a verdict on the lattice. The scalar ladder — one quantity in one dimension, pairs of quantities in two — is starved above its first rung. Two wide same-object axes are rare; three, which a 3D scalar engine would demand, do not exist in reachable public archives at all. Climbing that ladder further would mean building instruments with nothing to measure. So we do not climb it. We step off it.

Every engine the programme has built — phase and period, in one dimension and two, across fourteen volumes — has asked exactly one kind of question: *how large is a quantity, and does its magnitude land on the lattice?* Transition frequencies, velocities, masses, half-lives, luminosities: all of them scalars, all of them sizes. Not once has any engine asked where a thing *points*.

But a great deal of physical data is not a size at all. It is a **vector** — it has a magnitude *and* a direction. A star's motion points somewhere on the sky. A galaxy's spin has an axis. Light has a polarization angle. A crystal has an orientation. These directions are measured, by the million, in public archives — and no instrument in this programme has ever asked whether the lattice structures them.

**The Geometry**

**The new question, stated once:** does the geometry of the container leave a fingerprint on the *directions* of physical vectors — do they cluster toward preferred orientations more than an isotropic universe would allow — and are those preferred orientations the ones the 24-cell names?
This is the first question in the programme's history that engages the 24-cell as *geometry* — as an actual set of directions in space — rather than as the number 24 dividing a logarithm.

This volume does four things, and reports no result, because by the programme's own law no result may precede its pre-registration. First, it explains why the dimensional ladder ends and a new axis of inquiry begins. Second, it builds the mathematics of a vector engine from directional statistics and the geometry of the 24-cell, showing every formula in the open. Third, it sets out exactly how the vector engine differs from all four engines before it — 1D phase, 1D period, 2D

phase, 2D period — and what new falsifier the directional question requires. Fourth, it names the targets we will point it at, with their sizes and their hazards, and files the pre-registered expectations against which the built engine will be judged.

**Methodological Note**

**One inversion is worth stating at the outset, because it changes the character of the work.** The scalar engines were starved for data: the whole drama of Volumes 13 and 14 was dynamic range too narrow to resolve a register. The vector engine has the opposite problem. Directional data is abundant — millions of vectors, not forty-three nuclides. The limit is no longer statistical starvation; it is *systematic anisotropy*: survey footprints, the Galactic plane, dust, and selection all make directions cluster for mundane reasons that have nothing to do with the lattice. The vector engine's hardest job is not finding enough data. It is telling a lattice fingerprint apart from the shape of the telescope's field of view. That is a different discipline, and this volume pre-registers it.

— Timothy John Kish, Autumn 2026

# Chapter 8

# The End of the Ladder, and the Turn to Direction

## 8.1 Where the Scalar Ladder Stops

The programme's engines have climbed a ladder of dimension. It is worth drawing the whole ladder once, because its shape is the argument for stepping off it.

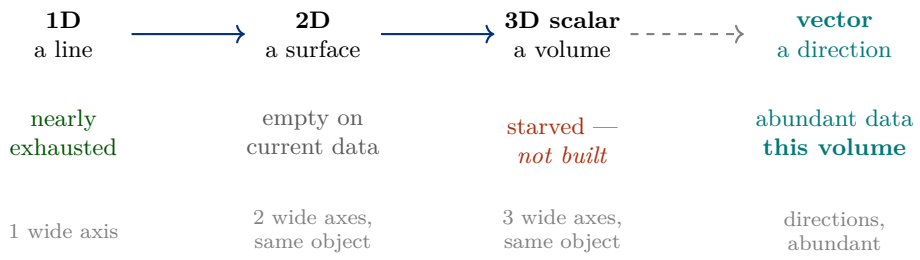



Figure 8.1: The dimensional ladder. Each scalar rung demands one more wide axis on the same objects, and each is harder to feed than the last: the 1D survey is nearly exhausted, the 2D survey is empty on current data (Volume 14), and a 3D scalar engine would need three wide same-object axes that do not exist. The vector turn is not a further rung — it is a step sideways, onto a question with abundant data of its own.

The arithmetic of the ladder is unforgiving. A 2D test needs two coordinates each spanning enough dynamic range to resolve a register, $M/N > 2.67$, measured on the *same* objects. Volume 14 found exactly one pair that reached even mid registers, and it returned a null. A 3D test needs three such axes at once, against a look-elsewhere space of $23^3 = 12{,}167$ register triples and a common-support floor compounding as $0.73^3 \approx 39\%$. If two-wide pairs are already down to zero in the public data, three-wide triples are not a smaller number — they are the empty set.

**Confirmed Signal**

**The justification bar for any new engine:** it must unlock a question the existing engines cannot ask, and it must have data it can actually be fed. The 2D turn cleared that bar — it unlocked the scramble null, the first falsifier able to tell joint structure from coincident marginals. A 3D scalar engine clears neither half: it asks the same magnitude question with more axes, and it has no data. The vector engine clears both. It asks a genuinely new question — about direction, which no prior engine touches — and directional data is abundant. That is why the ladder ends here and the compass begins.

## 8.2 What "Vector" Adds That "Scalar" Cannot Hold

A scalar is a single number: it lives on a line, and the only question one can ask of it is *where on the line*. A vector is a number *and* an arrow: it lives in a space, and it carries two independent pieces of information — how long it is, and which way it points.

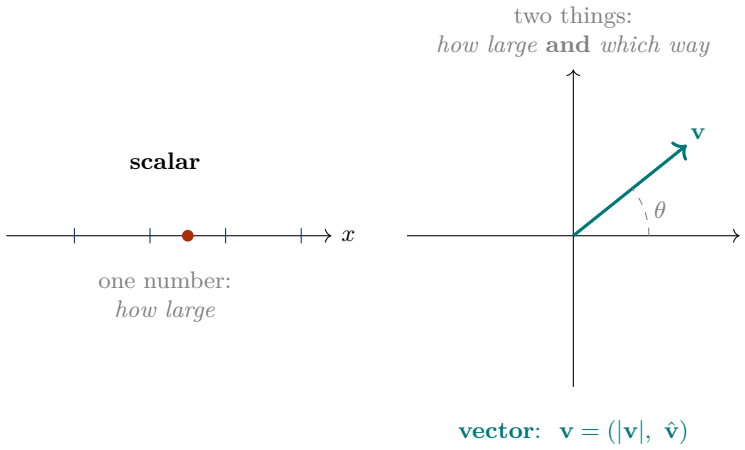



Figure 8.2: A scalar carries one fact: its size. A vector carries two, which separate cleanly — a magnitude $|\mathbf{v}|$ (a scalar, and the only thing the old engines could ever see) and a direction $\hat{\mathbf{v}} = \mathbf{v}/|\mathbf{v}|$ (a point on a sphere, which no engine has yet tested).

The split is the whole opportunity. The magnitude of a vector is a scalar, and the programme already has two mature engines for scalars — phase and period. Nothing new is needed there. The *direction* is a point on a sphere, and it is untouched territory. The vector engine is, at its heart, the pairing of the old magnitude question with a new direction question — and the coupling between them.

# Chapter 9

# The 24-Cell as an Orientation Lattice

*For fourteen volumes the number 24 has been a divisor in a logarithm — the container count that sets the register spacing. In this chapter it becomes something it has never been in this programme: a set of actual directions in space. This is the geometric heart of the vector engine, and it is why the engine is a genuine extension of the lattice idea and not merely a new statistic.*

## 9.1 What the 24-Cell Is

The 24-cell is the regular convex 4-polytope with 24 vertices, 96 edges, 96 triangular faces, and 24 octahedral cells. It is self-dual, vertex-transitive, and the unique regular polytope outside the simplex family with no direct analogue in three dimensions [16]. Its symmetry group is the Weyl group $F_4$, of order 1152.

Its 24 vertices have two standard coordinate presentations, and both matter:

**The Geometry**

**The $D_4$ root form.** The 24 vertices are all permutations of $(\pm 1, \pm 1, 0, 0)$:

$$(\pm 1, \pm 1, 0, 0),\ (\pm 1, 0, \pm 1, 0),\ (\pm 1, 0, 0, \pm 1),\ (0, \pm 1, \pm 1, 0),\ (0, \pm 1, 0, \pm 1),\ (0, 0, \pm 1, \pm 1).$$

That is $\binom{4}{2} \times 2^2 = 6 \times 4 = 24$ vectors. They are the root system of $D_4$, and they *generate* the $D_4$ lattice — the property that singles the 24-cell out, more than its group order [17].

**The unit-quaternion form.** Equivalently, the 24 vertices are the unit quaternions of the *binary tetrahedral group* $2T$:

$$\{\pm 1,\ \pm i,\ \pm j,\ \pm k\} \quad (8), \qquad \tfrac{1}{2}(\pm 1 \pm i \pm j \pm k) \quad (16),$$

for $8 + 16 = 24$ unit quaternions. Because unit quaternions *are* 3D rotations, this form says the 24-cell is a set of 24 preferred **orientations** — a natural, finite, highly symmetric "alphabet" of ways an object can be turned.

## 9.2 Two Spaces, One Polytope

Here is the honest subtlety that shapes the entire engine, stated plainly because hiding it would be exactly the kind of thing this programme refuses to do.

**Self-Correction**

**The 24-cell lives in four dimensions. Most directional data lives in three.** A star's velocity points somewhere on the ordinary 2-sphere $S^2$; the 24-cell's vertices are points on the 3-sphere $S^3$. They are not the same space. So the vector engine must be explicit about which test it is running, and it runs *two*, never conflating them:

1. **The native $S^3$ test (rotational data).** When the data is itself orientational — a full 3D attitude, a crystallographic orientation, a quaternion — it lives on $S^3$, and it is tested against the 24 vertices *directly*. This is the geometrically exact test: the 24-cell's own vertices are the preferred points. Crystallography is the natural home, because crystal orientations are routinely recorded as rotations and the cubic point groups are close relatives of $2T$.

2. **The projected $S^2$ test (ordinary direction data).** When the data is an ordinary 3D direction — a velocity, a spin axis, a line of sight — it lives on $S^2$, and it is tested against the *projection* of the 24-cell into three dimensions. The 24-cell projects naturally onto the **cuboctahedron** (12 vertices / directions) and its dual the **rhombic dodecahedron**; these 12 axes are the 3D shadow of the 24-cell's structure. The test asks whether 3D directions cluster toward those 12 axes beyond isotropy.

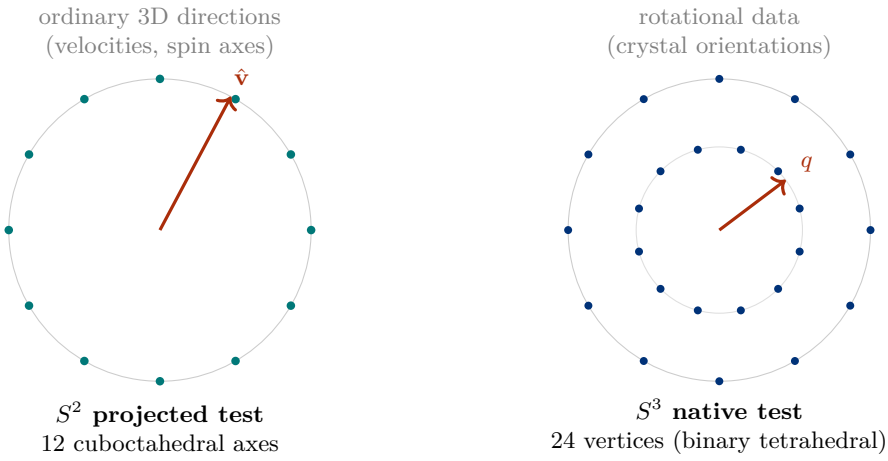



Figure 9.1: Two tests, one polytope, drawn schematically (the spheres are shown as circles). Ordinary 3D direction data is tested against the 24-cell's 3D shadow — the 12 cuboctahedral axes on $S^2$. Rotational data, which naturally lives on $S^3$, is tested against the 24 vertices themselves. The engine never mixes the two, and every result is labelled with the space it was scored in.

**Confirmed Signal**

**Why this is the right kind of audacious.** For the first time, the "24" in KishLattice is asked to do geometric work — to name directions, not to divide a logarithm. If the container's geometry is real, orientations should remember it. If it is not, they will be isotropic, and the engine will say so. Either way, the test finally engages the polytope as a polytope. That is a question worth building an engine for, and it is one no scalar engine could ever pose.

# Chapter 10

# The Statistic: Directional Resonance

*The engine has two channels, mirroring the two facts a vector carries. The magnitude channel is old — the existing phase and period engines, unchanged. The direction channel is new, and this chapter builds it from directional statistics in full view.*

## 10.1 The Magnitude Channel (Reused, Not Reinvented)

Every vector $\mathbf{v}_j$ has a magnitude $|\mathbf{v}_j|$, a positive scalar. On magnitudes the programme already has two mature, fingerprinted engines. The period engine asks whether $\ln|\mathbf{v}|$ repeats with a register period; the phase engine asks whether it sits near a register node. Nothing about the vector turn changes those statistics, and by the immutability principle the vector engine calls them rather than reimplementing them. The magnitude channel is settled territory.

## 10.2 The Direction Channel: Concentration on a Sphere

The direction $\hat{\mathbf{v}}_j = \mathbf{v}_j/|\mathbf{v}_j|$ is a unit vector — a point on the sphere. The question is whether a cloud of such points is *concentrated* toward the lattice axes, or spread *isotropically*. Directional statistics gives the tools exactly [10, 11].

**Derivation**

**The mean resultant.** For $n$ unit vectors, form the vector mean and its length:

$$\bar{\mathbf{R}} = \frac{1}{n}\sum_{j=1}^{n} \hat{\mathbf{v}}_j, \qquad R = |\bar{\mathbf{R}}| \in [0,1].$$

$R = 0$ is perfect isotropy (the arrows cancel); $R \to 1$ is perfect concentration (they align). This is the sphere's analogue of the circular resultant on which the period engine was built [12].

**The Rayleigh test for isotropy.** Under the null that directions are uniform on $S^2$, the statistic

$$Z = 3\,n\,R^2$$

follows a $\chi^2$ distribution with 3 degrees of freedom. A large $Z$ rejects isotropy. The resultant alone, however, only detects a single preferred direction — a *dipole*. The lattice predicts something richer: clustering toward *many* axes at once.

**Derivation**

**The orientation tensor (for multi-axis and headless data).** Spin axes and polarizations are *axial*: $\hat{\mathbf{v}}$ and $-\hat{\mathbf{v}}$ are the same datum, so the resultant $\bar{\mathbf{R}}$ cancels to zero and is useless. The correct tool is the $3 \times 3$ orientation tensor

$$\mathbf{T} = \frac{1}{n}\sum_{j=1}^{n} \hat{\mathbf{v}}_j\,\hat{\mathbf{v}}_j^{\top}, \qquad \operatorname{tr}\mathbf{T} = 1,$$

with eigenvalues $\tau_1 \geq \tau_2 \geq \tau_3$, $\sum \tau_i = 1$. Isotropy gives $\tau_1 = \tau_2 = \tau_3 = \frac{1}{3}$. A single-axis cluster drives $\tau_1 \to 1$; a girdle (plane) drives $\tau_3 \to 0$. The Bingham and Watson distributions [10] turn the eigenvalue spread into a calibrated significance.

## 10.3 The Lattice-Alignment Score

Isotropy tests ask only "are these directions non-random?" The lattice asks a sharper question: "are they non-random *in the specific way the 24-cell names*?" So the engine's headline statistic is an alignment to the lattice axes, not a generic anisotropy.

**Derivation**

Let $\{\mathbf{n}_k\}_{k=1}^{K}$ be the lattice axes ($K = 12$ cuboctahedral axes on $S^2$, or $K = 24$ vertices on $S^3$). For each datum define its alignment to the nearest lattice axis,

$$a_j = \max_k \left|\, \hat{\mathbf{v}}_j \cdot \mathbf{n}_k \,\right| \in [0,1],$$

and score the sample by the mean alignment $A = \frac{1}{n}\sum_j a_j$, reported always against the isotropic expectation $A_{\text{iso}}$ (the same quantity for uniformly random directions). The lattice-alignment excess is $A - A_{\text{iso}}$, and — exactly as in every prior engine — the number that is *ruled on* is not $A$ itself but its ratio to what the null produces on the same geometry.

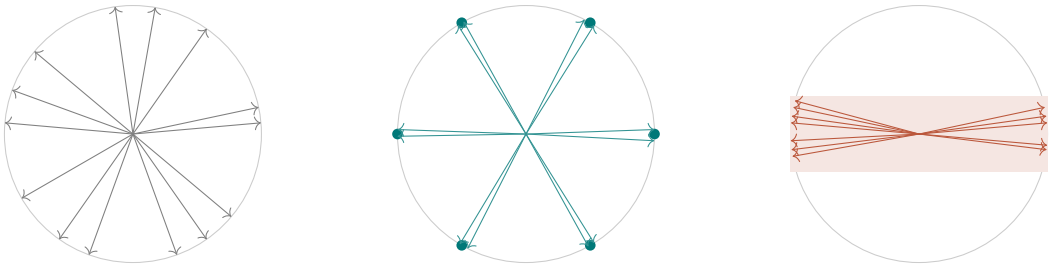



Figure 10.1: Three skies. Isotropic directions score at the null. Lattice-aligned directions pile toward the preferred axes and score well above it. But a survey that only sees a band of sky (right) also piles — for a mundane reason that has nothing to do with the lattice. Telling the middle panel from the right panel is the whole problem the next chapter's nulls exist to solve.

## 10.4 The Vector Lock: Coupling the Two Channels

The magnitude channel and the direction channel are each informative alone, but the lattice's deepest claim is that they are *coupled* — that a vector whose magnitude lands on a register also points along a lattice axis. A true **vector lock** is that joint event:

$$\underbrace{\big(\ \ln|\mathbf{v}_j| \text{ near register } N\ \big)}_{\text{magnitude channel}} \ \wedge\ \underbrace{\big(\, a_j > \text{threshold, toward axis } k\,\big)}_{\text{direction channel}}.$$

The vector lock is to the vector engine what the joint lock was to the 2D engine — and, as there, the danger is a spurious coupling produced by something other than the lattice. The instrument that disarms it is, once again, the scramble.

# Chapter 11

# The Nulls: Isotropy, Rotation, and the Selection Hazard

*The 2D engine taught the lesson that a null which can be fooled is worse than no null. The chaos null was fooled by single-axis structure; only the scramble caught it. The vector engine inherits exactly this shape of problem, and its nulls are built to the same discipline: one null to measure concentration, one null that cannot be fooled by the shape of the survey.*

## 11.1 The Isotropic Null

The directional analogue of the chaos null: directions drawn uniformly at random on the sphere, matched in number to the data, generated by normalising independent Gaussian vectors (the standard isotropic construction). It answers the first question — are the real directions more concentrated than random? — and it is the baseline $A_{\text{iso}}$ against which the alignment excess is measured.

**Artifact Form**

**But the isotropic null can be fooled, exactly as the chaos null was.** A survey does not see the whole sky. It sees a footprint — a band along the Galactic plane avoided, a hemisphere favoured, a dust mask cut out. Directions drawn from such a footprint are *not* isotropic, and they will clear the isotropic null handily — not because the lattice structured them, but because the telescope did. This is the directional form of the Form B hazard: structure that is real but mundane, masquerading as a lattice signal. An engine armed only with the isotropic null would announce the shape of the survey as a discovery.

## 11.2 The Scramble Null — the Spine, in Direction

As in every engine since the 2D turn, the scramble is the spine, and it is always on. In the vector engine it takes the coupling apart:

**Derivation**

**The magnitude↔direction scramble.** Shuffle which magnitude is attached to which direction, across the sample. This destroys any coupling between how large a vector is and where it points, while preserving *both* marginals exactly — the distribution of magnitudes is untouched, and so is the distribution of directions, footprint and all.

The consequence is precisely the one that redeemed the scramble in two dimensions. A survey footprint lives entirely in the direction marginal; the scramble preserves it. So if an apparent vector lock is really just "this anisotropic footprint happens to overlap the lattice axes," the scramble leaves it fully intact and the real-to-scramble ratio sits at unity. Only a *genuine coupling* — magnitudes on registers that also point along lattice axes, bound together — is broken by the shuffle. The scramble cannot be fooled by the shape of the sky, because the shape of the sky survives it.

## 11.3 The Rotation Null — Is the Frame Special?

The vector engine adds one null the scalar engines never needed, because direction has something magnitude does not: a *frame*. The lattice names specific axes; a real lattice signal should be aligned to *those* axes, not merely concentrated in some arbitrary direction.

**Derivation**

**The random-rotation null.** Apply a uniformly random global rotation $\mathbf{Q} \in SO(3)$ (or $SO(4)$ on $S^3$) to the entire set of lattice axes $\{\mathbf{n}_k\}$, leaving the data fixed, and recompute the alignment $A$. Repeat over many random frames to build the distribution of $A$ under "the lattice axes point nowhere in particular." A real signal clears this null: the data aligns to the *true* lattice frame far better than to a random one. A generic anisotropy (a dipole, a footprint) does not care which frame you pick, so it does not clear it.

**Confirmed Signal**

**Three nulls, three questions, by design.**

**Isotropic null** — is the sky non-random at all? (Can be fooled by the survey footprint.)

**Rotation null** — is the alignment specific to the *lattice* frame, not just any frame?

**Scramble null** — is the magnitude↔direction coupling real, or an overlap of independent marginals? (The spine. Cannot be fooled by the footprint.)

A vector lock counts only when it clears all three. The isotropic null establishes there is something to explain; the rotation null establishes the lattice frame is special; the scramble establishes the coupling is genuine. Any one alone is foolable; the three together are the test.

# Chapter 12

# How the Vector Engine Differs From All Four Before It

The programme now has four engines, and this is the fifth. The difference is not incremental — it is the first engine to leave the number line entirely.

| | **1D Phase** | **1D Period** | **2D Phase** | **2D Period** | **Vector** |
|---|---|---|---|---|---|
| **asks** | is a size near a node? | does a size repeat? | does a size *pair* sit on a node? | does a size pair repeat jointly? | does a *direction* align, and couple to size? |
| **space** | line | line | plane | torus | sphere ($S^2$ / $S^3$) |
| **data** | scalar | scalar | scalar pair | scalar pair | **vector** |
| **statistic** | node residual | circular resultant | joint cell | toroidal resultant | resultant / orientation tensor / axis alignment |
| **uses "24" as** | divisor | divisor | divisor | divisor | **geometry (vertices/axes)** |
| **nulls** | chaos | annular | chaos + scramble | annular + scramble | **isotropic + rotation + scramble** |
| **data appetite** | 1 wide axis | 1 wide axis | 2 wide axes, same obj. | 2 wide axes, same obj. | **directions — abundant** |

Table 12.1: The five engines. The vector engine is the first to leave the line for the sphere, the first whose datum is a vector rather than a scalar, the first to use the number 24 as a set of directions rather than a divisor, and the first whose data is plentiful rather than scarce. It adds two nulls the scalar engines never needed — the rotation null (a frame question) and the isotropic null (a concentration question) — while keeping the scramble as the spine.

**Methodological Note**

**The continuity that matters.** For all its novelty, the vector engine keeps the three commitments that define every KishLattice instrument. The scramble null is the spine and is never tunable off. The resolution and significance criteria are re-derived for the new space before any data is scored. And every gate must be shown to FAIL on a fixture built to fail it before it is trusted to PASS. The engine is new; the discipline is not.

# Chapter 13

# Targets: What We Will Point It At

*Numbers up front, as always. The vector engine's targets are chosen for abundant, well-characterised directional data and for one more thing the scalar targets never had: each is a standing question in its own field about whether directions are truly isotropic — the exact question a null-first directional auditor is built to answer.*

| **Target** | **source** | $n$ | **space** | **directional observable / hazard** |
|---|---|---|---|---|
| Stellar space velocities | Gaia DR3 | $\sim 10^6$ | $S^2$ | 3D velocity direction; hazard: disk/footprint anisotropy |
| Galaxy spin axes | SDSS, DESI | $\sim 10^5$ | $S^2$ axial | spin-axis alignment (a standing, contested claim) |
| CMB polarization angles | Planck | map | $S^2$ axial | low-$\ell$ alignment (the "axis of evil" anomaly) |
| Pulsar spin–velocity | ATNF | $\sim 10^2$ | $S^2$ | known kick alignment — a positive-control class |
| Crystal orientations | materials DBs | $\sim 10^5$ | $S^3$ | native 24-cell test; hazard: preferred growth axes |

Table 13.1: Candidate vector targets. Unlike the scalar survey, the limiting quantity is not sample size — it is systematic anisotropy, named in the last column for each. The pulsar spin–velocity alignment is listed deliberately as a *known* real alignment: a class the engine must recover, or it is not trustworthy on the cases where the answer is unknown.

### The Geometry

**Why these, and in what order.** The pulsar spin–velocity alignment is a real, published, non-lattice alignment — so it is the field's gift of a positive control: a genuine directional signal the engine must detect to prove it can. The CMB low-multipole alignment (the "axis of evil") is a standing anomaly in cosmology that has never been weighed by a null-first directional auditor built against selection and look-elsewhere from the start — precisely our kind of question. Galaxy spin alignment is a decades-old, still-contested claim, drowning in selection concerns — again our kind of question. Crystal orientations are the one target that engages the 24-cell *natively* on $S^3$, with abundant data and a clean frame. Gaia velocities are the largest and cleanest sample, and the natural first light on real data.

### Methodological Note

**The honest expectation, filed in advance as always.** The vector engine may find nothing — just as the scalar engines mostly found nothing. Physical directions may be isotropic with respect to the 24-cell, the rotation null may show no special frame, and the scramble may destroy every apparent coupling. That is a possible and acceptable outcome, and because the data is abundant it will be a *clean* null — a real verdict, not the resolution-limited shrug that the starved scalar pairs could only give. If directions do cluster on the lattice axes and survive all three nulls, it will be the first thing the programme has found that no scalar engine could reach, and it will have engaged the 24-cell as geometry for the first time. Either way, the answer is earned.

# Chapter 14

# Pre-Registrations for the Vector Engine

Filed before a line of engine code, under the programme's standing law. The built engine will be judged against this list, and any departure from it will be reported as such.

**Formal Pre-Registration**

**Standing pre-registrations, Volume 15.**

1. **Immutable from the scalar engines.** The direction channel is new code with its own fingerprint; the magnitude channel *calls* the existing phase/period engines unchanged.
2. **The scramble null (magnitude↔direction) wired in from the first commit**, as the primary coupling falsifier, always on, never tunable off.
3. **Three nulls required for any vector lock:** isotropic (concentration), rotation (frame), and scramble (coupling). A lock is reported only when it clears all three; clearing fewer is reported as exactly that.
4. **The $S^2$ and $S^3$ tests never conflated.** Every result is labelled with the space it was scored in and the axis set used (12 cuboctahedral axes, or 24 vertices).
5. **Axial data handled with axial statistics.** Headless directions (spins, polarizations) are scored with the orientation tensor / Bingham, never with the resultant, which vanishes for axial data.
6. **The selection function modelled or masked before scoring.** Survey footprint, Galactic-plane cut, and dust mask are declared per target and carried into the isotropic null, so that the baseline is the *observed* sky, not an idealised full sphere.
7. **A directional resolution/power criterion re-derived and published** before scoring: the sample size needed to detect a given alignment excess above the isotropic floor, which scales as $1/\sqrt{n}$. The analogue of $M/N$, stated for each target in advance.
8. **Look-elsewhere corrected for the axis count and frame search**, analogous to the $23 \times 23$ correction of the 2D engine.
9. **A pre-registered positive control:** the pulsar spin–velocity alignment, which the engine must recover, filed as a required pass before any unknown target is trusted.
10. **Synthetic calibration before real data:** a lattice-aligned set (must clear all three nulls), an isotropic set (must null), and a *selection-striped* set planted to look aligned and required to be rejected by the scramble — the directional expected-fail.
11. **Every gate demonstrated to return fail on a fixture built to fail it**, before it is trusted to PASS.
12. **Visual inspection of sky maps and sphere plots is part of the verification chain**, not a presentation step — the directional analogue of the 2D heatmap rule.

# Chapter 15

# What We Hope To See — and What We Will Have Either Way

**Confirmed Signal**

If directions cluster on the 24-cell's axes and survive the isotropic, rotation, and scramble nulls together, it will be the first positive the programme has ever produced that no scalar engine could reach — and the first evidence that engages the container as a geometric object rather than a number in a formula. It would be extraordinary, and it would have passed the hardest test the programme knows how to build.

**Confirmed Null**

And if they do not — if physical orientations are isotropic with respect to the lattice, the frame is not special, and the scramble destroys every coupling — then we will have established that cleanly, on abundant data, with a real verdict rather than a resolution-limited silence. We will say so as plainly as we have said every null before it. A clean null on a million vectors is worth more than a hopeful signal on forty-three.

Either way, the programme ends this volume with something it did not have before: an instrument that reads *vectors*, not just scalars — the most adaptable tool for consuming data lakes the programme has built. Across fifteen volumes the toolkit has grown from a single-scalar comb to a matched pair of scalar engines, to a matched pair of surface engines, and now to an engine that measures direction itself. Whatever the 24-cell turns out to be, the instruments for asking are public, fingerprinted, and permanent — and each one can say no.




Figure 15.1: We built engines that measure how large things are. Now we build one that measures where they point. The container, if it is real, has a shape — and a compass is the instrument that feels for a shape in the directions of things. –Timothy John Kish

# References

## The KishLattice Series

All volumes are published open-access with full data, configuration, and engine fingerprints. Every pre-registration listed below was filed prior to the run it governs.


- Kish, T. J. (2026). *Holographic Resonance: The Geometry of a Quantized Universe* (Vol. 1). Zenodo. https://doi.org/10.5281/zenodo.18209530
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  *Direct predecessor. Pre-registers both lake rebuilds closed in Chapters 1–2 of the present volume, and carries the resolution criterion $M/N > 2.67$ used throughout.*


## Pre-Registrations and Supporting Papers


- Kish, T. J. et al. (2026). *Noise Does Not Lock — A Comprehensive Empirical Response to Skeptical Challenges to KishLattice Geometric Harmonic Spectroscopy.* Zenodo. `https://doi.org/10.5281/zenodo.20585516`
  *Source of the Grid-Phase Locking Theorem and of the scramble-null analysis (Erratum 1) on which Chapters 4–6 turn.*

- Kish, T. J. et al. (2026). *Master Equation of the Kish Lattice Theory — The Rosetta Atlas.* Zenodo. `https://doi.org/10.5281/zenodo.18235735`
  *Source of the lattice energy expression and of the definition of $f$ carried forward from Volume 13.*

- Kish, T. J. et al. (2026). *Volume 13 — Pre-Registration P13.1–P13.6.* Zenodo. *Carries the resolution census and the two rebuild specifications upheld in Chapters 1–2.*

- Internal memorandum AM-15-01 (2026). *Vol 15 Direction: The General Falsifier — Bode's Law, the Exoplanet Hand-off, and the $k_{\text{geo}}$ Refactor; the Dimensional Ladder and the Vector Engine.* KishLattice 16$\pi$ Initiative LLC.
  *Records the go/no-go discipline of the nuclear probe in Chapter 7 and the dimensional-ladder reasoning deferring the 3D scalar engine in favour of the vector engine.*


## Repository and Project Resources

- Kish, T. J. *Holographic Resonance: The Geometry of a Quantized Universe* — engine source, configurations, audit scripts, and run receipts. GitHub. `https://github.com/TimothyKish/Holographic-Resonance-The-Geometry-of-a-Quantized-Universe`

- KishLattice 16$\pi$ Initiative LLC. `https://www.kishlattice.com`

The scripts released with this volume reproduce each table and verdict from the published lakes, and each aborts rather than substituting a value when a required field is absent:

- `kish_lake_io.py` — strict field contract; recursive resolver; coverage reporting.

- `build_s2_kinematics_rebuild.py`, `promote_rebuild.py` — the `s2` covariance rebuild of Chapter 1 ($5\sigma$ parallax and covariance-propagated proper-motion gates; DR3 zero-point; computed $M/N$).

- `periodogram2d.py`, `phase_lock2d.py` — the two 2D engines (Chapters 5–6); the scramble null is always on; the census compares best-of-grid real against best-of-grid scramble.

- `build_velocity_luminosity_lake.py`, `support2d.py` — the maiden-voyage lake and support audit of Chapter 6.

- `probe_nuclear_runway.py` — the go/no-go feasibility probe of Chapter 7; measures span and gate on each axis of the strict join and scores nothing.
- `make_synthetic_lakes.py` — the four permanent 2D control lakes (seed-fixed, regenerable byte-for-byte).

## Primary Data Sources — Vol 14

- **Gaia Collaboration** (2023). Gaia Data Release 3. VizieR catalogue I/355.
  *Source of the `s2` stellar-kinematics rebuild (Chapter 1) and of the velocity×luminosity lake (Chapter 6, $n = 83{,}155$ promoted pairs).*
- **LIGO/Virgo/KAGRA Collaboration**, Gravitational-Wave Transient Catalogue GWTC-3, with `pyRing` ringdown posteriors. `https://www.gw-openscience.org/`
  *Source of the `l1_gwtc` rebuild (Chapter 2); measured $f_{220}$, $n = 21$ Kerr-consistent events.*
- **NNDC / IAEA nuclear data** — NuDat and the NUBASE2020 evaluation. `https://www.nndc.bnl.gov/nudat3/`
  *Source of the `Q5` decay lake (half-life axis) of the Chapter 7 probe.*
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  *Source of the `Q4` nuclear lake (decay-energy axis) of the Chapter 7 probe; strict join yielded 43 matched nuclides.*
- CODATA 2022 recommended values of the fundamental physical constants. `https://physics.nist.gov/cuu/Constants/`

## Primary Data Sources — Vol 15 Targets

- **Gaia Collaboration** (2023). Gaia Data Release 3. VizieR catalogue I/355.
  *Stellar space-velocity directions; the largest and cleanest $S^2$ target.*
- **Sloan Digital Sky Survey / DESI.** `https://www.sdss.org/` ; `https://www.desi.lbl.gov/`
  *Galaxy spin-axis orientations; axial $S^2$ data for the spin-alignment question.*
- **Planck Collaboration** (2020). Planck 2018 results. *Astronomy & Astrophysics*, 641, A6.
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- **ATNF Pulsar Catalogue.** `https://www.atnf.csiro.au/research/pulsar/psrcat/`
  *Pulsar spin and proper-motion directions; the pre-registered positive control.*
- **Materials orientation databases** (e.g. MP, EBSD repositories).
  *Crystallographic orientations; the native $S^3$ test against the 24 vertices.*

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# Appendix A

# The Filter Ablation: What Each Guard Contributes to the Verdict

*The 2D engines carry three tunable form filters, built as toggles on purpose so that their contribution could be measured rather than assumed. This appendix measures it. We run the full factorial — every on/off combination of the three filters — against lakes with known answers, and report exactly how much each guard moves the verdict. The result is clean, a little surprising, and it closes the 2D engine with a number instead of a claim.*

## A.1 The Point of the Ablation

A filter that is always on is a filter whose contribution you cannot see. From the first commit the 2D engines exposed their artifact filters as switches — `common_support`, `scale_flag`, and the `resolution_gate` — precisely so that we could run the engine with each one on, then off, and read the difference directly off the census. The question this appendix answers is empirical and specific: *how much does each form-artifact guard actually contribute to the final verdict, and do any of them compound?*

**Methodological Note**

**Which guards are toggles, and which are not.** Of the programme's five artifact forms, only three correspond to a tunable filter. The scramble null — the defence against Form B (mundane correlation) and single-axis structure — is the *spine*: always on, never a toggle, and therefore not ablated. Its contribution is read in *every* cell of the experiment as the real-to-scramble ratio itself. Form A (the transform artifact) is the open gate (`FORM_A_UNCLEARED`), a theory question, not a filter. So the factorial is over exactly the three tunable guards.

| **toggle (default on; `-no-X` off)** | **targets** | **role** |
|---|---|---|
| `common_support` | Form D — null-support / floor mismatch | per-coordinate support correction |
| `scale_flag` | Form C — scale degeneracy / preprocessing | per-axis rescaling flag |
| `resolution_gate` | the $M/N > 2.67$ criterion | refuses unresolvable register pairs |
| *scramble null* | *Form B + single-axis* | *the spine — always on, not a toggle* |

Table A.1: The three tunable guards and the always-on spine. The ablation toggles the top three; the scramble is read in every run.

## A.2 The Design

Three binary toggles give $2^3 = 8$ combinations per engine. The full factorial captures each filter's individual effect *and* every pairwise and triple interaction, so compounding is visible if it exists. Each combination was scored, through the identical pipeline, against the four synthetic control lakes (where the ground truth is known by construction) and the real velocity×luminosity lake.

| `common_support` | `scale_flag` | `resolution_gate` | reads |
|---|---|---|---|
| on | on | on | baseline (what Chapters 5–6 reported) |
| off | on | on | Form D alone removed |
| on | off | on | Form C alone removed |
| on | on | off | resolution gate alone removed |
| off | off | on | C + D compounding |
| off | on | off | D + gate compounding |
| on | off | off | C + gate compounding |
| off | off | off | all guards removed — raw statistic |

Table A.2: The $2^3$ factorial. The harness `ablate_2d.py` drives the real pipeline as a subprocess under each combination, with the scramble null on throughout, and tabulates the best-of-grid real-to-scramble ratio per lake.

## A.3 The Results

Numbers up front, exactly as returned. The striking feature is visible before any interpretation: the ratio column does not move.

| **2D PERIOD engine** | | | | **best-of-grid ratio** | | |
|---|---|---|---|---|---|---|
| combo | CS | SF | RG | `jointlock` | `noise` | `vel×lum` |
| 111 | Y | Y | Y | 16.91 | 1.09 | 0.88 |
| 110 | Y | Y | – | 16.91 | 1.09 | 0.88 |
| 101 | Y | – | Y | 16.91 | 1.09 | 0.88 |
| 011 | – | Y | Y | 16.91 | 1.09 | 0.88 |
| 100 | Y | – | – | 16.91 | 1.09 | 0.88 |
| 010 | – | Y | – | 16.91 | 1.09 | 0.88 |
| 001 | – | – | Y | 16.91 | 1.09 | 0.88 |
| 000 | – | – | – | 16.91 | 1.09 | 0.88 |
| **spread across all 8:** | | | | **0.00** | **0.00** | **0.00** |

Table A.3: Period engine ablation. The best-of-grid ratio is identical in all eight combinations, to the precision reported. CS = common_support, SF = scale_flag, RG = resolution_gate.

| **2D PHASE engine** | | | | **best-of-grid ratio** | | | | |
|---|---|---|---|---|---|---|---|---|
| combo | CS | SF | RG | `formB` | `jointlock` | `noise` | `oneaxis` | `vel×lum` |
| 111 | Y | Y | Y | 1.00 | 2.24 | 0.97 | 0.98 | 1.00 |
| *all other 7 combos* | | | | 1.00 | 2.24 | 0.97 | 0.98 | 1.00 |
| **spread across all 8:** | | | | **0.00** | **0.00** | **0.00** | **0.00** | **0.00** |

Table A.4: Phase engine ablation. Every lake returns an identical ratio in all eight combinations. The planted lock survives (2.24); the correlation ridge, pure noise, single-axis artifact, and the real lake all sit at the floor — and none of that depends on the tunable filters.

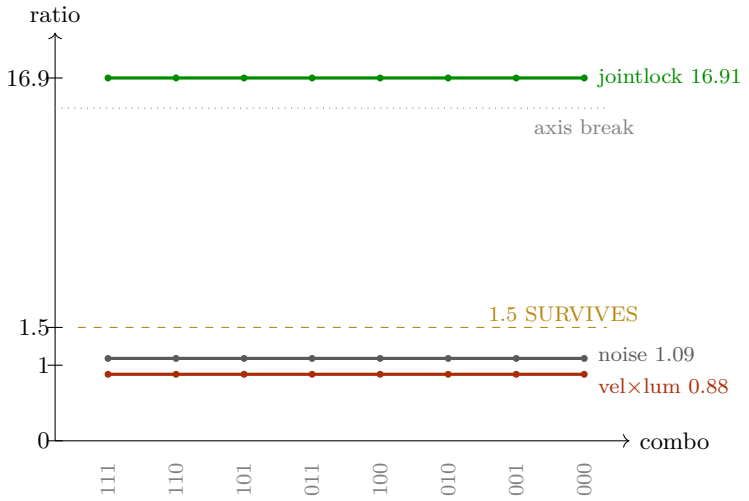



Figure A.1: Verdict invariance (period engine; the phase engine is identical in character). Every lake's best-of-grid ratio is a dead-flat line across all eight toggle combinations. The planted lock sits far above the SURVIVES threshold; the nulls and the real lake sit at the floor; and nothing — not a single guard, nor any compounding of them — moves any line. The tunable filters do not change the verdict.

## A.4 Finding 1 — The Verdict Is Invariant; the Scramble Carries It

The headline is the flatness. Across all eight combinations, in both engines, every lake returns the same ratio it returned with every guard on. Removing Form D, removing Form C, removing the resolution gate, and removing all three at once change the census verdict by 0.00.

**Confirmed Signal**

**The scramble null carries the entire verdict.** The three tunable filters contribute nothing to the final number on these lakes. The one defence that is *not* a toggle — the scramble, the spine — is the one that determines every outcome. This is the strongest possible statement about a falsifier's integrity: the verdict rests on the single component that can never be switched off, and is robust to every knob that can.

There is a clean reason for this, and it is worth writing down because it is structural, not accidental.

**Derivation**

The census statistic is a *ratio* of two best-of-grid powers,

$$\rho \;=\; \frac{\max\limits_{(N_1,N_2)\in\mathcal{G}} P_{\text{real}}(N_1,N_2)}{\max\limits_{(N_1,N_2)\in\mathcal{G}} P_{\text{scr}}(N_1,N_2)}.$$

A filter acts in one of two ways: the resolution gate restricts the eligible grid, $\mathcal{G} \to \mathcal{G}_f$; the support and scale corrections reweight the per-pair power, $P \to P_f$. Crucially, every filter is applied *identically to the real lake and to its scramble*, because the scramble passes through the entire pipeline as a sibling lake. So both numerator and denominator are transformed by the same $f$:

$$\rho_f \;=\; \frac{\max\limits_{\mathcal{G}_f} P_{f,\text{real}}}{\max\limits_{\mathcal{G}_f} P_{f,\text{scr}}}.$$

When the winning pair survives the filter for both real and scramble — which it does whenever the winner is already a well-resolved, well-supported pair — the correction multiplies top and bottom by the same factor and the ratio is unchanged. **The filters move the floor; they do not move the ratio of the two peaks above it.** The ablation confirms the cancellation empirically: $\Delta\rho = 0.00$.

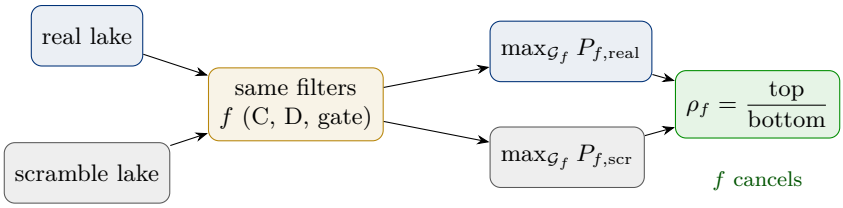



Figure A.2: Why the filters cancel. The same filter $f$ is applied to the real lake and its scramble before the ratio is taken. A correction that multiplies the real peak also multiplies the scramble peak, so it divides out. The verdict is, by construction, robust to the tunable guards — and determined by the scramble.

## A.5 Finding 2 — The Resolution Gate Governs Coverage, Not Verdict

One toggle does change an observable quantity — but not the one that decides anything.

| lake | resolution gate | pairs scored | ratio |
|---|---|---|---|
| `synth` lakes | on | 225/529 | unchanged |
| `synth` lakes | off | 529/529 | unchanged |
| `velocity×luminosity` | on | 112/529 | 0.88 |
| `velocity×luminosity` | off | **529/529** | **0.88** |

Table A.5: The resolution gate's only effect is coverage. Turning it off scores every pair on the grid; the best-of-grid ratio is unchanged, because the pairs the gate removes were never going to win.

**Confirmed Signal**

**This strengthens the velocity×luminosity null of Chapter 6.** With the gate on, only 112 of 529 pairs were scoreable — and a skeptic could fairly ask whether the null was an artifact of the gate quietly hiding the other 417. The ablation answers that directly: with the gate *off*, all 529 pairs are scored, and the ratio is still 0.88. The null is not a gate artifact. It holds across the entire register grid. The gate's role is honesty of coverage — it refuses to report pairs the data cannot resolve — not suppression of a signal.

## A.6 Finding 3 — The Honest Boundary: C and D Had Nothing to Bite On

The temptation here is to over-read Finding 1 — to conclude the Form-C and Form-D filters are useless. That conclusion would be wrong, and stating why is the honest close.

**Self-Correction**

**Zero contribution here is not zero contribution in general.** The `common_support` and `scale_flag` filters are built to suppress a *specific* artifact each: a floor-mismatch concentration (Form D) and a scale-degeneracy alignment (Form C). None of the four control lakes *contains* such an artifact reaching the best-of-grid. The planted set is a true joint lock, pure noise, a Form-B correlation ridge, and a single-axis artifact — and the scramble already disposes of the last two (`formB` 1.00, `oneaxis` 0.98). The C and D filters were handed nothing to catch, so their measured contribution is correctly zero. This is a statement about the *lakes*, not a proof the filters never matter.

To test the C and D filters' protective value properly, the engine must be shown a lake that contains the very artifact each is built to suppress — and shown that the filter, and only the filter, removes the spurious lock. That is a clean, bounded follow-up, and we file it as a pre-registration rather than leave it implied.

**Formal Pre-Registration**

**Pre-registration (filed for the next 2D calibration pass).**

1. **A Form-C planted control:** a lake with a phase-neutral rescaling of one axis chosen to manufacture a joint alignment. Required result: a spurious lock that *survives the scramble* but is removed by `scale_flag` — the case that would prove the filter catches something the spine does not.
2. **A Form-D planted control:** a lake with a large below-floor population on each axis, so the clamp manufactures near-node mass. Required result: a spurious lock removed by `common_support` applied per-coordinate.
3. If either filter fails to remove its planted artifact, or if the scramble already removes it (making the filter redundant), that is itself the reportable finding.

## A.7 What the Ablation Closes

**Confirmed Null**

**The empirical standing of the 2D guards, measured not asserted.** On the lakes tested — four controls with known answers and the one real pair — the census verdict is invariant to all three tunable filters ($\Delta\rho = 0.00$, both engines, all eight combinations). The scramble null, the one component that is never a toggle, carries the verdict entirely. The resolution gate governs coverage, not outcome, and the velocity×luminosity null holds with the gate off across the full 529-pair grid. The Form-C and Form-D filters are confirmed inert on the current lakes, for want of a planted artifact to suppress; their protective value is pre-registered for a dedicated calibration pass.

This is the way the programme closes every component: with a number that could have come out otherwise. The filters *could* have moved the verdict, and we would have reported by how much. They did not, and we report that — along with the honest reason, and the test that would

complete the picture. The 2D engine's tunable guards are now characterised, and the spine is shown, empirically, to be exactly that.

*We built the filters as switches so their worth could be weighed, not assumed. Weighed, they move the verdict by nothing — and in moving nothing, they prove that everything rests on the one thing we never let anyone switch off.*

# Appendix B

# Formal Verification with Lean: Certifying the Instrument

*This appendix is a pre-registration, filed in Volume 14 ahead of Volume 15. It commits the programme to formally verifying the* mathematics of the instrument *in the Lean proof assistant — not the physics of the lattice, which only data and nulls can decide. It states the firewall between those two things plainly, lists the exact theorems we will prove, and shows the shape of one of them in Lean so the commitment is concrete rather than aspirational.*

## B.1 Why Formal Verification, and Why Now

Across fourteen volumes the programme has made a particular kind of promise: that it hides nothing. The lakes are public, the engine source is public, every gate is computed rather than printed, and every run leaves a fingerprint. Formal verification is the next rung of that same ladder, and it closes the one gap the others cannot: the gap between *a proof we wrote in prose* and *a proof that is actually correct*.

A prose proof asks the reader to trust the author. A formal proof asks the reader to trust nothing but a tiny, fixed logical kernel that mechanically checks every step. There is no alien math, no hidden equation, no hand-waving between lines — if the proof compiles, the stated theorem genuinely follows from the stated assumptions, and any human can read the statement, audit it, and back-check the reasoning for themselves. This is precisely the standard the recent wave of formalized mathematics has adopted: a result is accompanied by machine-checkable code so the community verifies the *proof*, not the *reputation* of whoever announced it.

**Methodological Note**

**The point in one line.** A fingerprint certifies *which* code ran. A Lean proof certifies that *what the code claims to compute is mathematically what it computes.* The first stops a silent substitution; the second stops a subtle error in the statistic itself.

## B.2 The Firewall: Lean Certifies the Instrument, Not the Physics

This is the most important sentence in the appendix, and we put it before anything else Lean can do.

**Methodological Note**

**Lean verifies the instrument. It cannot verify the lattice.** A machine-checked proof that the period engine is unit-invariant, or that the scramble is inert in one dimension, says nothing whatever about whether $k_{\text{geo}} = 16/\pi$ structures reality. That is an empirical question, settled by data and by nulls, and no proof touches it. Lean certifies the *ruler*; only measurement certifies the *thing measured*. If the word "verified" is ever allowed to drift from "the instrument is correctly specified" to "the hypothesis is true," the programme has broken its own first rule.

This is the same boundary first light drew for the 2D engines, restated one level deeper. First light proved the instrument behaved honestly on data with known answers; Lean proves the instrument is *mathematically* what we said it is, before any data at all. Both are statements about the tool. Neither is a statement about the universe.

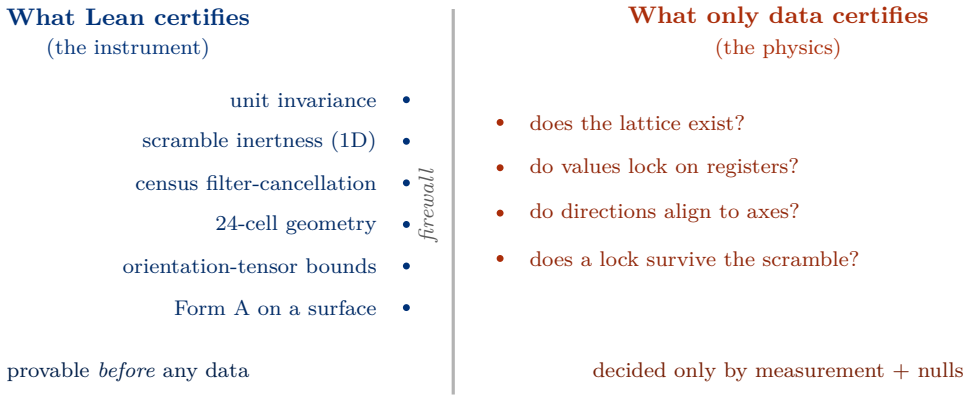



Figure B.1: The firewall. Everything on the left is a mathematical property of the instrument and is provable, in principle, with no data at all — this is Lean's domain. Everything on the right is an empirical property of the world and is decided only by data and the nulls. Lean can never cross the wall, and the programme never lets "verified" pretend it has.

## B.3 How Lean Works, Briefly

Lean is an open-source functional programming language and interactive theorem prover, built around a deliberately *tiny* trusted kernel: every proof, however it was produced, is ultimately reduced to steps that small kernel checks. Its mathematical library, Mathlib, contains millions of lines of formalized mathematics — groups, root systems, quaternions, linear algebra, measure theory — that a new proof can build on. The same machinery underwrites automated-reasoning efforts at the major AI labs, where machine-checkable proof is the scoring standard precisely because it cannot be faked.

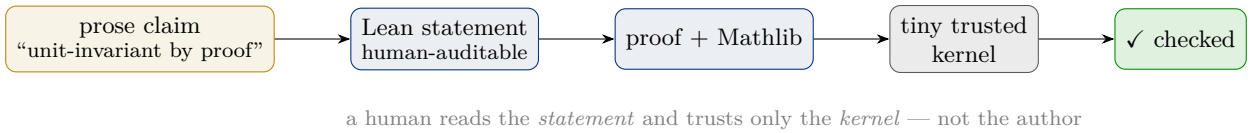



Figure B.2: From claim to check. A prose claim becomes a precise Lean statement a human can read and agree is the right thing to prove; the proof is then reduced, through Mathlib, to steps a tiny fixed kernel verifies. If it compiles, the theorem holds. Reputation plays no part.

**Methodological Note**

**A fitting coincidence.** Lean 4's build tool is named **Lake** (“Lean Make”); its projects are configured by a `lakefile`. We have built *lakes* of data for fifteen volumes; the proof layer, it turns out, builds with Lake too. The deeper fit is real, not just the name: Lean's whole design is a *tiny trusted core that checks everything else* — the same principle as our fingerprints, our strict input contract, and “the engine decides the truth.”

## B.4 The Pre-Registered Target Theorems

Filed before a line of Lean is written. These are the obligations the proof module must discharge, and any that resist — most pointedly Form A — will be reported as open exactly as they stand.

**Formal Pre-Registration**

**Lean verification targets (VL-1 through VL-7), filed in Volume 14.**

1. **Period-engine unit invariance.** For the period resultant $R$ and any scale $c > 0$,

$$R(\{c\,x_j\}) = R(\{x_j\}),$$

because $\ln(cx) = \ln c + \ln x$ adds a constant to every phase — a global rotation the resultant magnitude cannot see.

2. **Erratum 1 — scramble inertness in 1D.** For any statistic $f$ that reads only the *multiset* of scalars, and any permutation $\sigma$,

$$f(\sigma(x)) = f(x).$$

The one-dimensional scramble is therefore inert by construction.

3. **Census filter-cancellation.** For the census ratio $\rho = \max_{\mathcal{G}} P_{\text{real}} / \max_{\mathcal{G}} P_{\text{scr}}$, a filter applied identically to real and scramble that preserves the argmax leaves $\rho$ unchanged (the result measured empirically in Appendix A).

4. **The 24-cell, as geometry.** The 24 vertices equal the $D_4$ root system and form the binary tetrahedral group of order 24; the symmetry group $F_4$ has order 1152; the projection to $S^2$ yields the 12 cuboctahedral axes. (Mathlib carries quaternions, finite groups, and root systems; these are among the most tractable targets.)

5. **Orientation-tensor bounds.** The tensor $\mathbf{T} = \frac{1}{n}\sum_j \hat{\mathbf{v}}_j \hat{\mathbf{v}}_j^{\top}$ is symmetric positive-semidefinite with $\operatorname{tr}\mathbf{T} = 1$, eigenvalues $\tau_i \in [0, 1]$, and isotropy $\Rightarrow \tau_1 = \tau_2 = \tau_3 = \frac{1}{3}$.

6. **Alignment-score properties.** The lattice alignment $a_j = \max_k |\hat{\mathbf{v}}_j \cdot \mathbf{n}_k| \in [0, 1]$ is rotation-equivariant: $a(\mathbf{Q}\hat{\mathbf{v}};\, \mathbf{Q}\mathbf{n}) = a(\hat{\mathbf{v}};\, \mathbf{n})$ for any rotation $\mathbf{Q}$.

7. **Form A on a surface (flagship, open).** State the one-dimensional transform-artifact theorem formally, then prove or refute its extension to a surface. A Lean proof would close the `FORM_A_UNCLEARED` gate with machine-checked certainty — stronger than any prose or model-assisted derivation.

**Methodological Note**

**Order of attack.** VL-1 through VL-3 are the invariance quartet's core — short proofs that harden claims already in print, and the right first commits. VL-4 through VL-6 verify the vector engine's foundations *before* the engine is built, which is the pre-registration law applied to the proof layer. VL-7, Form A, is the prize, and it is allowed to stay open as long as it must.

## B.5 The Shape of One Theorem

To make the commitment concrete rather than a promise, here is Erratum 1 (VL-2) in Lean. The idea is small and exact: a statistic that reads only the *multiset* of scalars cannot see order, so a

permutation of the sample — the scramble — cannot move it.

```
-- Erratum 1: a lock statistic that reads only the MULTISET of scalars
-- is blind to order, so any scramble (a permutation of the sample)
-- leaves it unchanged -- the scramble is inert in one dimension.

structure LockStat where
  f : Multiset ℝ → ℝ            -- the statistic sees the multiset, not the order

theorem scramble_inert_1d
    (S : LockStat) (xs σ : List ℝ) (h : σ ~ xs) :   -- σ is a permutation of xs
    S.f ↑σ = S.f ↑xs :=
  congrArg S.f (Multiset.coe_eq_coe.mpr h)          -- equal multisets ⟩ equal value
```

Listing B.1: VL-2, the shape of it. Schematic Lean 4; Mathlib lemma names may need adjustment on first compile.

**Derivation**

What the proof says, in words: the hypothesis `h` states that $\sigma$ is a permutation of the sample. Casting both lists to multisets (the $\uparrow$ coercion) erases order, so the two multisets are *equal*; applying the statistic to equal inputs gives equal outputs. Three lines, and the kernel checks every one. The same discipline applied to VL-1 would leave a single open goal marked `sorry` — Lean's honest placeholder for "not yet proved" — which is exactly how a pre-registered-but-unproved obligation should appear on the record.

## B.6 Scope and Honest Limits

**Self-Correction**

**What Lean will and will not reach, stated in advance.**

- **Strong where we start.** Mathlib's algebra and geometry are deep, so the invariance quartet (VL-1–3) and the 24-cell facts (VL-4) are genuinely tractable.
- **Thin where the distributions live.** The asymptotic distributions — the Rayleigh $3nR^2 \sim \chi^2_3$, the Bingham and Watson laws — are heavy to formalize and may not exist in Mathlib. We do not pretend those will be verified soon; we scope to the algebraic and geometric statements first and say so.
- **Verification, not discovery.** Lean certifies what we have reasoned; it will not find new physics, and it cannot rescue a null. The data still rules.
- **The firewall again.** Nothing Lean proves is evidence for the lattice. Every verified theorem is a statement about the instrument.

## B.7 What This Commits Us To

**Formal Pre-Registration**

A Lean 4 proof module, built with Lake and depending on Mathlib, will live in the project repository beside the Python engines, fingerprinted like everything else. It will discharge VL-1 through VL-6 as they are proved, carry VL-7 (Form A) openly until it is closed, and mark every unproved obligation with `sorry` so the state of the proof is never misrepresented. The module certifies the instrument; the lakes and the nulls, as always, are what speak to the physics.

*We built engines that measure, and nulls that can say no. Now we add a layer that lets anyone check the mathematics of the measuring itself — not by trusting us, but by trusting a kernel that trusts no one. The ruler will be provably straight. Whether it finds the lattice is still, and always, for the data to say.*

## Sources for this Appendix


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- Google DeepMind (2024). *AlphaProof*: formal mathematical reasoning with Lean. `https://deepmind.google/` *(Representative of machine-checked proof as the scoring standard in automated reasoning.)*




There is no Noise only Data. Remove the filters and see the Universe raw.